

**Using Gamification To Influence User Success
In Personal Finance Applications**

by

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in partial fulfilment of the
requirements for the degree of
Master of Digital Media
in the program of
Digital Media

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Abstract

Research Problem & Objectives: This thesis explores the current research and prior literature surrounding financial literacy's use as a tool for consumer empowerment. It seeks to understand the current market for financial literacy digital media in the form of mobile budgeting applications. In particular, to understand where weaknesses exist and opportunities lie.

Methods: That will entail combining the existing literature and best practices with persuasive technology (gamification) to create a minimum viable product that may address deficiencies in the market. This proposed application will be designed to be an intervention to increase the end user's level of financial literacy and financial well being.

Key Results: This study resulted in the development of a new mobile personal finance budgeting application uniquely designed to make financial literacy information more accessible and available to users, and to help users improve their financial habits & behaviours. This was accomplished through the implementation of gamification techniques.

Conclusion: This thesis concluded having created a tool that Canadians might use to increase their level of financial literacy. The mobile budgeting app developed filled a gap in the Canadian mobile personal finance application market by employing research-backed gamification techniques, multiple engagement loops and behavioural change strategies.

Keywords

gamification, fintech, personal finance, financial well being, persuasive technology, budgeting, financial literacy, user engagement, education, F.I.R.E.^{[P]_[SEP]}

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Table Of Contents

Abstract.....	iii
Keywords.....	iv
Acknowledgements.....	v
Table Of Contents	8
List Of Tables.....	10
List Of Figures	11
List Of Appendices	13
1.0 Thesis Statement	14
2.0 Contributions	15
3.0 Introduction	16
4.0 Review of Related Literature	18
4.1 History of Financial Literacy Education	18
4.2 Financial Literacy Education’s Efficacy.....	19
4.3 Finlit Education In Digital Media.....	20
4.4 Financial Behaviours Through The Lens of Behavioural Psychology	23
4.5 Gamification: A Tool For Behavioural Change	24
4.6 Fintech: A Playground For Financial Education & Behavioural Change.....	25
4.7 A Case Study For Change: The F.I.R.E. Movement	26
4.8 Opportunities	28
5.0 Design & Methodology	29
5.1 Project Goals.....	29
5.2 Review of Related Literature.....	29
5.3 Market Survey	31
5.4 Product Development	33
5.5 Reflection.....	34
6.0 Results	35

6.1 Market Survey of Canadian Mobile Budgeting Applications	35
6.1.1 Introduction	35
6.1.2 Overview of Application Features	35
6.1.3 Gamified Features	36
6.1.4 Analysis	42
6.2 Recommendations	44
6.3 Prototype Design	46
6.3.1 Introduction	46
6.3.2 Onboarding.....	46
6.3.3 Skills.....	55
6.3.4 Progress	61
6.3.5 Additional Sections	65
6.4 Prototype Development	69
6.5 Takeaways.....	77
7.0 Conclusions	78
7.1 Conclusions	78
7.2 Future Work	78
Appendices.....	80
References.....	91

List Of Tables

Table 1 - Analysis of applications through the lens of Yu-Kai Chou's Octalysis framework.	41
Table 2 - Analysis of applications using Brian Cugelman's Behaviour Change Strategies.	41

List Of Figures

Figure 1 - Anthropomorphism of app in onboarding.	46
Figure 2 - Introduction to life energy in onboarding.	48
Figure 3 - Card deck swiper used to define user's dreams.	50
Figure 4 - Anthropomorphism providing mentorship in onboarding.	52
Figure 5 - Introduction of rewards in onboarding.	53
Figure 6 - Extreme customization feature in onboarding.	54
Figure 7 - Linear layout of skills development path.	57
Figure 8 - Introduction page in skills area of the app.	58
Figure 9 - User flow of a skills challenge which demonstrates user choice.	59
Figure 10 - Seedling & sun reward modals within the app.	60
Figure 11 - Rewards page within the app.	61
Figure 12 - Customizable budget categories.	66
Figure 13 - Transaction entry screen.	67
Figure 14 - Research notes.	70
Figure 15 - Paper Prototype of Mobile Budgeting Application.	70
Figure 16 - Research notes grouped by theme.	71
Figure 17 - Areas of greatest motivational value identified.	71
Figure 18 - Areas of least motivational value identified.	72
Figure 19 - Low-code draft of application's basic pages.	72
Figure 20 - Sign up, login, help pages & navigation.	73
Figure 21 - Application database design.	73
Figure 22 - Review & synthesis of research notes for individual pages.	74
Figure 23 - Final sketches of app pages & gamified features.	74
Figure 24 - Draft of game narratives.	75
Figure 25 - Game narrative draft on low-code platform.	75
Figure 26 - Rewards modals.	76

Figure 27 - Application user flow.	76
Figure 28 - Application multimedia elements.	77

List Of Appendices

Appendix A - Cognitive Biases.....	80
Appendix B - Excluded Mobile Budgeting Applications	81
Appendix C - Yu-Kai Chou’s Octalysis Framework.....	83
Appendix D - Brian Cugelman's Persuasive Architecture of Gamification.....	85
Appendix E - Financial Consumer Agency of Canada’s Research Priorities	86
Appendix F - Complete List of Prototype Application’s Features.....	88

1.0 Thesis Statement

This paper posits that mobile fintech apps which are currently available in the Apple App store and Google Play store in Canada could more effectively engage and build motivation for consumers so they are retained as long-term users. It hypothesizes that incorporating gamification techniques will make financial literacy education and mobile fintech apps more engaging to users and increase their motivation to use them, thus increasing their ability to derive benefit from them.

Through an exploration of the current research, case studies and prior literature surrounding personal finance and financial literacy's use as a tool for consumer empowerment, this paper will establish that financial literacy efforts have had a positive effect on the financial outcomes of participants. It will go further to show that delivery of financial literacy efforts through digital media correlate with increased positive financial outcomes. Finally, this paper will establish that the inclusion of gamified techniques has lead to an increase in motivation and willingness to engage with, and ability to derive benefit from fintech applications.

It will conclude by discussing which gamification techniques are currently being employed by popular Canadian fintech applications and where opportunities lie for the inclusion of potentially more effective gamification techniques. The project will make use of digital media and persuasive technology to develop a concept application designed to address those opportunities. The concept will be built as a partially functioning app using no-code techniques. That will be followed by a discussion of the design rationale and creation process for the prototype, along with recommendations for future research. [P]
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2.0 Contributions

This project adds to the current body of research by illuminating strategies fintech budgeting application companies might employ that have the potential to increase motivation, engagement and positive outcomes for their users. Applications that are currently available on the market could be doing a better job of accomplishing this. There is potential to incorporate gamification techniques to help users achieve better financial outcomes. It furthers the conversation and research being done in areas where financial literacy and persuasive technologies intersect. This research is important because according to Menard (2017), “Financial stress among employees is reaching epidemic proportions: 75% live paycheck to paycheck, personal savings rates are at their lowest since 2007, and non-mortgage debt levels are higher now than during the Great Recession.” According to the Financial Consumer Agency of Canada (2020), there is plenty of room for new research on consumer financial protection as well as the development of technologies to contribute to improvements in financial education.

3.0 Introduction

Canadians are facing increased pressure as the wealth gap widens. During the pandemic, Canada's richest 44 people got \$53 billion wealthier (Tencer, 2020). Meanwhile, Canada's unemployment rate increased from 5.67% in 2019 to 9.75% in 2020 (Statista, 2020). A survey conducted by CIBC found that 38% of respondents have borrowed money to cover day-to-day expenses during the pandemic (Burgess, 2020). The number of months to save a downpayment for a home in Canada has also reached a record high. The median household needs 60 months of savings for the minimum downpayment and it is as high as 409 months (34 years) in places like Toronto and Vancouver (Punwasi, 2021).

Moving out of their parent's homes, buying a house and starting a family is becoming an increasingly unrealistic dream for many millennials (Bleemer, 2014). Meanwhile, the gap between Canada's highest wage earners and their lowest paid employees is growing. According to the Chartered Professional Accountants of Canada (2021):

“Rising income inequality continues to be an economic, social and political concern both internationally and here in Canada. Inequality may have only risen modestly, on average, across the country, but economic forces continue to concentrate both income and wealth among Canada's richest. This represents a serious concern for our country's standard of living.”

Many can't help but feel frustrated reading news stories about government corruption and corporate missteps that are increasingly being paid for by retail consumers, while those responsible escape consequences. The recent r/wallstreetbets-lead rise of GameStop's (GME)

stock price in response to excessive shorting by hedge funds is testament to this growing sentiment (Chohan, 2021).

Many Canadians are feeling discouraged and disempowered by growing inequality. With wage growth falling in relation to inflation, consumers are faced with an increasingly urgent and inelastic problem (Press Progress, 2019). This paper will explore how principles of gamification can make a meaningful difference for consumers by increasing their motivation within, and engagement with financial literacy and personal-finance-focused smartphone apps. It will go on to demonstrate a minimum viable product designed to address the current gaps in the personal finance app market, which by filling, may help consumers to increase their level of financial literacy and financial well being. [P]
[SEP]

4.0 Review of Related Literature

This work builds on a number of previous studies on the effectiveness of financial literacy education and gamification techniques. Financial literacy refers to a combination of financial awareness, knowledge, skills, attitudes and behaviours necessary to make sound financial decisions and ultimately achieve individual financial well-being (FCAC, 2020; CFPB, 2015; Muir et al., 2017; FCAC, 2019).

4.1 History of Financial Literacy Education

Financial literacy education has been informal throughout much of human history. The first appearance of financial advice in literature came in the form of a quote by Benjamin Franklin in 1737 in which he wrote “A penny saved is two pence cleared; A pin a day’s a groat a year.” It was included in an article titled “Necessary Hints To Those That Would Be Rich” that was later compiled into an almanac and published by Houghton, Mifflin & Company (1886).

In the 19th century, according to Chris Arthur, “Like the financially literate individual, consumer financial literacy education did not appear ex nihilo, but instead builds upon its neoclassical economic and consumer education foundations.” He writes, “Neoclassical economics, solidifying into a discipline with the writings of William Stanley Jevons, Carl Menger, Leon Walras and Alfred Marshall in the nineteenth century, transformed the discipline of economics and produced a new vision of the economic sphere that was much narrower than its predecessor, political economy.” (Arthur, 2012). An early example of formalized financial literacy education can be found in James William Gilbert’s book *The Elements of Banking*

(1855). In it, he writes a chapter titled “Ten Minutes’ Advice About Keeping A Banker” for the layman in which he details the benefits of opening a bank account and provides the reader with instructions, noting things like, “If the sum be under 10%, or if the sum be above 10%, and you are not likely to want it soon, put it into the savings bank; you will receive interest for it at the rate of about 3% for every 100% for a year.” (Gilbart, 1855).

In North America, modern financial literacy education efforts began when, “in the US, the Smith-Lever Act of 1914 established the funding that land grant colleges still use to teach cooperative extension courses in personal finance.” (Menard, 2017). Menard notes that, “congress, along with nonprofit community organizations, continued to support these educational efforts over the decades and in 2003 established the Financial Literacy and Education Commission.” According to the Government of Canada (2020):

In the mid-1990s, financial education or financial literacy emerged as a key priority for many countries around the world. More and more government officials began to recognize the broader economic implications of their citizens’ poor financial knowledge and decision-making on their national economies, and how it could lead to significant global repercussions.

During this time, Canada’s government set in motion an extensive period of review and public consultations. These resulted in a series of reforms to Canada’s financial sector and the creation of the Financial Consumer Agency of Canada (Government of Canada, 2020).

4.2 Financial Literacy Education’s Efficacy

Several studies have established that financial literacy education can be effective. In 2001, Clancy, M. et al. found that attendees of a financial literacy education program had a higher average monthly deposit number and deposited more frequently than those who had not attended the program. Further, the average monthly deposit amount increased as the total number of hours the participants spent in the program increased.

In a study of workplace financial education conducted by Bernheim and Garrett (2003), the median savings rates were found to be 22% higher for individuals whose employers offered financial education, versus those whose employers did not. This study accounted for savings that were separate from workplace saving and retirement plans (Fox, et al 2005).

In another study on the efficacy of financial literacy education, Bernheim et al (2001) studied students who attended schools in a U.S. state where financial literacy education was mandated and compared them to students who attended schools in a state in which it was not. They found evidence of the positive effect of financial education on savings rates and net worth for those students during their peak earning years, age 35–49.

Financial literacy education has grown from its infancy in the 18th century to become a priority for Canadian government and non-governmental organizations as a response to deteriorating social security within the context of rising inequality. With numerous scientific studies backing its efficacy, we can establish that financial literacy education has produced positive financial outcomes for participants of these programs.

4.3 Finlit Education In Digital Media

Concurrent with an increase in consumer financial literacy education efforts, there were technological advances which resulted in the emergence of the digital media field. The emergence of digital media laid the groundwork for the creation of the fintech industry. Gulamhuseinwala et al. (2015) define fintech businesses as, “firms that are combining innovative business models and technology to enable, enhance and disrupt financial services.” With the increasing popularity of cloud based online financial tools and mobile-based financial planning apps, consumers are progressively adopting financial technology to help manage their personal finance matters (Bi, et al., 2020; Bi & Huston, 2014).

Following this movement, organizations offering financial literacy education have increasingly adopted digital tools to deliver their content. Literature unanimously points toward the advantages of adopting technology and its implications for learners, as well as teachers and the teaching pedagogical system (Rasco, et al., 2020; Bang & Dalsgaard, 2006; Berti, 2016; Mayer, 2002; Siu & Koh, 2016; Smith & MacGregor, 1992; Sorensen & Murchú, 2006). Several studies detail how the intersection of financial literacy education and digital media has been beneficial for consumers.

A study by Li, J. And Meyer-Cirkel, A. (2019) involving high school students was able to demonstrate the effectiveness of using a digital platform to increase financial literacy. Using a digital financial literacy tool called *saalt*, students were asked to compose a balanced financial portfolio. They then gradually observed changes in asset prices over time. Observing the sample portfolio provided by the *saalt* platform, students were able to adjust their portfolio weights using the information they observed. The study also showed that students were influenced by

portfolios shared amongst their classmates on Facebook. Both digital influences allowed students to adjust their portfolios to produce better financial results.

A study of college students by Bi, Q. et al. (2020), found that that financial software use serves as a complement to financial literacy education in improving the individual's financial knowledge. The study administered financial literacy questionnaires to participants before, and four weeks after they were given a one-day financial literacy workshop. Respondents who used fintech software both before and after the workshop had the highest levels of financial knowledge during the study. The study reflected on these findings, saying:

Financial software use may be helpful to make respondents monitor their financial situation, visualize how their current savings impact their future retirement lifestyle, or make complicated calculations easier for individuals and lead to better financial decisions such as saving for retirement (Bi et al., 2017).

After analyzing a 2001 Survey of Consumer Finances, Hogarth and Anguelov (2004) found that consumers who use e-banking technologies are likely to engage in more positive financial behaviours (Bi, et al., 2020). They found:

Households that used PC banking had a probability of being in the “better” (financial) manager group that was 8 basis points higher compared with those that did not.

Households that used direct deposit had a probability of being in the “better” (financial) manager group that was 7 points higher than others (Hogarth & Angeulov, 2004).

These studies show that financial literacy education maintains its efficacy, and may even be enhanced when delivered in a digital format. Further, these results suggest that simply having

interacted with fintech products in the past correlates with better financial outcomes for participants.

4.4 Financial Behaviours Through The Lens of Behavioural Psychology

While it has been established that financial literacy education can be effective, some researchers have pointed out its limitations. A comprehensive survey conducted by Willis, L. (2009) points to inconclusive studies and problematic methodologies when asserting that it's difficult to establish whether an increase in financial knowledge leads to improved financial behaviours. She suggests that improved financial outcomes could be incidental, or related to factors not accounted for in the scientific studies reviewed. While this study and others are not cause to discard the results of previous studies on the efficacy of financial literacy education altogether, it does suggest new areas of research that could lead to improvements.

One of those areas might be behavioural economics. Menard, M. (2017) points out that the study of behavioural economics might enhance our ability to deduce the motivations behind, then be more effective at changing, the actions of consumers as it relates to their personal finances. In the 1970's Daniel Kahneman and Amos Tversky published papers that introduced the idea of behavioural economics. They combined cognitive psychology with classic economic analysis to coin the term "cognitive biases" to explain the imprudent decisions people make after evaluating financial choices. For a list of these cognitive biases, see Appendix A.

Cognitive biases explain why people sometimes behave less than rationally when it comes to their money and finances. Even when people know what they should do with their money, they still make choices that run counter to their best interests. This makes a compelling

argument for why financial literacy education efforts should go beyond simply increasing, then testing a participant's level of financial knowledge. These papers by Kahneman and Tversky suggests that directly influencing a participant's financial behaviours might be just as, if not more important than, influencing their base level of financial knowledge.

It is in this area of financial literacy education, as well as in attracting and retaining users of financial literacy and personal finance apps, that we find an ideal opportunity to introduce gamification.

4.5 Gamification: A Tool For Behavioural Change

Gamification is defined as the use of game design elements in non-game contexts (Deterding, et al., 2011). Yu-Kai Chou (2014) adds to this definition, describing gamification as human-focused design, which is “a design process that remembers the human motivations within the system.” B.J. Fogg (2019), describes *motivation* as a fundamental element in creating behavioural change, alongside having the *ability* to make said change, and experiencing a *prompt* to do so. This suggests that a professional practice which is defined as paying attention to human motivations when designing systems for entertainment, engagement and learning, might be ideal for creating behavioural change. Both Willis' assertion that financial knowledge does not necessarily lead to behaviours that will generate improved financial outcomes and Kahneman and Tversky's papers indicating that we possess biases which cause us to behave in ways that are counter to our financial best interests suggest placing a focus on behavioural change when discussing ways to improve financial literacy education and personal finance apps could be fruitful.

Gamification's potential to modify human behaviours has been well studied. The presence of game elements evokes experiences (Hamari et al., 2014) and promotes intrinsic motivations to engage in activities (Hamari and Koivisto, 2015). These motivations can result in greater user engagement (Kwon et al., 2015; Penkkimäki et al., 2015; Shivnetra, 2017), enhanced customer loyalty (Saidon et al., 2016), improved effectiveness and efficiency of learning (Deterding et al., 2011) and even concrete behavioural changes (Rayuk and Altobello, 2018; Maturo and Setiffi, 2015; Mitchell et al., 2017; Washburn, 2017).

In a literature review of 24 peer reviewed papers which included empirical studies designed to answer the question "Does gamification work?", researchers found that in the majority of the studies reviewed, gamification did produce positive effects and benefits (Hamari, et al., 2014).

4.6 Fintech: A Playground For Financial Education & Behavioural Change

The fintech industry presents itself an ideal venue for financial literacy education efforts to take place if one is attempting to effect behavioural change. Research shows that a carefully designed mobile fintech app with gamified features has the potential to help consumers improve their level of financial literacy and financial well being. A study by Rodrigues, et al. (2016) collected data that found that bank customers responded positively to gamified banking applications. Participants highlighted the ease-of-use and revealed a strong intention to use the application.

In a separate study of gamification's impact on the acceptance of mobile banking services, it was found that its effect on use behaviour was significant. Gamification positively

impacted and significantly influenced behavioural intention, confirming the importance that the use of game mechanics and game design techniques can have on the intention to use mobile banking services (Oliviera & Baptista, 2016). Previous studies by Venkatesh et al. (2003, 2012) found that the path coefficient of behaviour intention on use behaviour was significant, suggesting that having an intention to use something means the user is significantly more likely to do so (Baptista & Oliviera, 2016). Gamification in these studies, created an openness to, and greater likelihood that consumers would use the fintech app being studied.

In a combined study of gamification and financial literacy education, an American non-profit organization named Commonwealth created a game called *SavingsQuest* to help users save money for an emergency fund. They conducted a study of its efficacy and found, “in our pilot with a prepaid card company, *SavingsQuest* users saved on average 25 percent more frequently than other cardholders” (Maynard and McGlazer, 2017).

These are just a few examples that demonstrate the ability gamification has to increase a user’s willingness to engage with, ability to derive benefit from, and financial results as a result of the use of fintech apps.

4.7 A Case Study For Change: The F.I.R.E. Movement

F.I.R.E. is an acronym that stands for “financial independence, retire early”. It’s a lifestyle movement focused on financial independence that began with Vicki Robin’s 1992 book *Your Money Or Your Life*. In it, she equated money to life energy. She pointed out that the working class exchange time in their lives (life energy) for money, and that those who do so

should carefully consider the money they spend, as it represents time they might not otherwise have to work. (Robin, 1992) In her book, she suggests a number of strategies that would help readers achieve financial independence before the age of 65. Her ethos became popular when she appeared on the Oprah Winfrey Show in 1992. The book became a New York Times best seller.

Through the late aughts of the 21st century, the movement crystallized into a series of tenets designed to help adherents retire before the expected retirement age of 65. Many adherents of the F.I.R.E. movement retire in their 30's or 40's. The central tenet of the movement is to aim for savings rate of 50-70% of one's annual income. Ramsey. (2021) Supporting tenets include reducing expenses, adopting a minimalist lifestyle, DIY investing and increasing one's income. Many F.I.R.E. adherents seek out high paying tech jobs and build multiple sources of income in order to achieve this target.

While the movement received limited mainstream media attention in 2018, it remains largely underground. Adherents follow it through popular blogs, YouTube channels and podcasts. A podcast by F.I.R.E. influencer Mr. Money Moustache is a popular source of information, as is influencer Nick True's YouTube channel. Other F.I.R.E. authors include Grant Sabatier and Jacob Lund Fisker. (Moss, 2018). F.I.R.E. is not well known within the general population, however a poll completed by TD Ameritrade found that 11% of wealthier Americans aged 45 and older had heard of the F.I.R.E. movement by name, while another 26% were aware of the concept. The Harris Poll (2018). A 2020 study by Taylor & Davies characterized the F.I.R.E. movement as consumers using "finance against capitalism".

It is the position of this paper that the intersection of financial literacy education, fintech and gamification can be strengthened by the addition of tenets borrowed from this ethos.

4.8 Opportunities

Based on the review of existing research, there is an opportunity to make financial literacy education more widely available and more accessible to Canadians through digital media, so that users may improve their level of financial knowledge.

There is an opportunity to help users of digital media tools overcome their cognitive biases through the implementation of gamification techniques, so that they might improve their financial behaviours. Combined, these interventions have the potential to improve the user's overall level of financial well being.

5.0 Design & Methodology

5.1 Project Goals

This research project explores persuasive technology for financial literacy as implemented through a digital app prototype that addresses gamification opportunities within the fintech mobile app field. This project was completed in the interest of making financial literacy education tools more widely available, accessible and effective, to become a potential source of consumer empowerment in the face of growing inequality in Canada. It was to be followed by the writing of a reflexive paper on the creation of a concept application.

5.2 Review of Related Literature

It began with a literature review focusing on the areas of financial literacy, fintech and gamification. The literature review surveyed existing academic research sourced using Google Scholar and the Ryerson University Library. It began with a chronological overview of financial literacy education, starting from its infancy to set the stage for a discussion of modern financial literacy education. Research which commented on the genesis, and progression of formal financial literacy education in the US was included to provide Western cultural context. Finally, research published by the Government of Canada was cited to provide local context, since this project focuses specifically on mobile personal finance apps which are available in Canada.

The review then covered studies which demonstrate that financial literacy education efforts delivered via analog means can be effective. It also included studies that show financial literacy education delivered via digital media can be effective. The studies that examined the

digital delivery of financial literacy education were selected for their recency and relevance to the subject of financial literacy.

The review also included studies which suggest financial literacy education efforts may not be effective in all cases and provide direction on areas for future research. This was done to ensure the literature review contained a fair and balanced accounting of all research relevant to the subject of financial literacy education, and contained multiple citations which suggest areas for improvement. Studies rooted in behavioural psychology were reviewed for this section, since the hypothesis of this paper is that improvements rooted in gamification, a subset of the behavioural psychology field, may prove to be viable solutions to the challenge of designing a more impactful financial literacy education tool.

Following that path, studies which examine the efficacy of gamification in creating behavioural change were surveyed. Well-cited studies, which form the foundation of the study of gamification and its efficacy, were selected for this section to provide a foundation for the discussion of gamification as it has been applied to financial literacy education. Studies which have examined the efficacy of digital financial literacy education tools that incorporate some gamified elements were reviewed. These were included to provide an overview of where the industry is currently sitting regarding the inclusion of gamification techniques and their efficacy when applied to financial literacy education. This discussion set the stage for a market survey that delved more deeply into popular and/or well-gamified financial literacy tools which are currently available in the Canadian mobile application market. Finally, a discussion of the F.I.R.E. movement and associated studies was included to frame a possible path towards financial success for users.

5.3 Market Survey

A market survey was designed to assess the most popular and/or well-gamified personal finance and budgeting apps available in Canada, and identify gaps that might be filled with a new budgeting app.

For the purposes of this review, “personal finance application” is defined as a personal budgeting application containing expense tracking and budgeting features. Within the fintech sphere, mobile banking applications and applications which primarily focus on investing were excluded from this review. They were excluded because their core functionality was not developed with the expressed intent of helping users improve their financial knowledge, skills and habits - collectively referred to as their level of financial literacy. (FCAC, 2020; CFPB, 2015; Muir et al., 2017; FCAC, 2019). Mobile banking and investing apps are primarily designed to provide their users with mobile digital access to their banking and investing accounts and the functionality that typically accompanies those products - namely, features like online bill payments features, enabling stock purchases and providing stock information.

The applications that were included in the review contained core functionality that was underpinned by their creators’ intent to help their users to, in some way, improve their level of financial literacy. While it’s unclear whether financial literacy education is the goal of all of the applications reviewed, they are the fintech sphere’s closest representation of the aforementioned effort. Most, if not all, imply in their advertising that an increase in their user’s level of financial well being might be a possible result of using their product.

A brief review of 51 budgeting applications which are available to Canadians on the Apple App Store and Google Play Store was completed to assess their inclusion of gamified

features. Twenty one budgeting applications were given a brief review and quickly eliminated due to their lack of budgeting functionality.. The remaining 30 applications were reviewed to determine the extent to which they had incorporated gamified features. Based on this criteria, a list of three personal finance and budgeting applications were selected for an in-depth review. They were selected for their either extensive or novel implementation of gamified features relative to other similar apps on the market. An additional two applications were chosen for in-depth review as the most widely downloaded personal finance and budgeting applications in Canada with over 10 million downloads each on the Google Play Store. A total of five budgeting applications were given an in-depth review. For a list of the 25 applications that were reviewed and excluded from the final analysis, see Appendix B. All five applications chosen for final analysis enjoy wide popularity in Canada. A minimum of 1 million downloads each was required for inclusion in this paper to ensure the applications reviewed were already impacting a substantial number of users and therefore, potentially substantially impacting the Canadian financial literacy landscape.

A detailed analysis of the five chosen applications was performed using both Yu-Kai Chou's Octalysis Framework (see Appendix C) and Brian Cugelman's Persuasive Architecture of Gamification (see Appendix D). These frameworks were sourced using Google Scholar and through professional networking. You Kai-Chou's framework was selected because it is a heavily cited framework describing the practice of gamification. Brian Cugelman's architecture was chosen to include the perspective of a Canadian behavioural psychology expert who specializes in gamification, since the project aims to help Canadians. The gamified features present in each app were catalogued using these two frameworks. The number of instances in

which each gamification tenet appeared in the reviewed apps were tabulated and presented in chart form.

5.4 Product Development

At the conclusion of the market survey, the gamification tenets that were absent or only appeared in low numbers relative to others were highlighted as gaps, and identified as areas where a greater inclusion of them in a new budgeting app may increase motivation and engagement for the user. This was presented as a list of recommendations for a new mobile budgeting app. These recommendations defined the core features of a minimum viable product which was designed to address these areas. A concept and mock-up was developed to walk the viewer through a new budgeting app experience which uses the principles of gamification to motivate and engage users, with the goal of having them derive an increased benefit from it as a financial literacy education tool.

The concept and mock-up was developed using the low-code platform Adalo. A low-code mobile application development platform was selected because it has the ability to demonstrate the designed application's visual, motion and interaction design elements, in addition to the application's information architecture. Adalo was selected because of its ability to develop applications for mobile, ease of use, the visual appeal of finished products on the platform, ability to adequately demonstrate gamification concepts and ability to perform live demonstrations of the application for external viewers. Twenty-six low and no-code platforms were evaluated using the above criteria. This was done using an Excel spreadsheet in which the

criteria were weighted and the results ranked before Adalo was selected. The list of 26 platforms was sourced through attending live webinars, networking meetings, professional contacts, mentorship, a search of low and no-code discussion forums, a search of low-code online publications and thorough a google search for any additional platforms missed during earlier searches.

5.5 Reflection

The project concluded with a results section discussing the prototype's features and their underlying design rationale. The inclusion of each feature was supported using Yu-Kai Chou's Octalysis framework, insights gleaned from research reviewed during the literature review and through a thematic analysis of the current literature on the subject. Yu-Kai Chou's framework was selected for this discussion because it is a heavily cited framework describing the practice of gamification.

User experience, user interface and classic design techniques were also employed in the development of the prototype, but were not discussed in the results section in the interest of constraining the scope of the paper to the theme of gamification.

The results section also illuminated the prototype creation process through a detailed descriptions of the steps taken when designing and building the app. Figures showing notes and sketches while the prototype was being created were included to illustrate the progression of the project from concept to finished prototype. Figures showing screen captures of the finished app while it was being demonstrated were included to show the finished product.

6.0 Results

6.1 Market Survey of Canadian Mobile Budgeting Applications

6.1.1 Introduction

In this market survey five mobile budgeting applications which are available to Canadians have been selected for in-depth review and analysis. The applications have been analyzed for their inclusion of gamification techniques. The five applications have been broken into two categories based on whether they provided users with the tools needed to change their financial behaviour in the future, or whether the app simply reported the user's past financial behaviours back to them. For the purposes of this paper, applications falling into the latter category are described as “passive”. Applications falling into the former category are described as “active”. Two frameworks have been chosen to categorize the gamified features included in the apps: Yu-Kai Chou’s Octalysis framework and Brian Cugelman’s Persuasive Architecture of Gamification. A summary of each framework can be found in Appendices C and D.

6.1.2 Overview of Application Features

Money Manager and Mint are the most widely downloaded personal finance applications on the Android and iOS platforms, both having in excess of 10 million downloads within Canada. They are both marketed as personal finance budgeting applications. Both apps allow users to track past spending and employ an envelope-style budgeting system which allows users to lay out categories and assign monthly budgets to them. Both apps are considered to be

passive, since they provide data to their users on their past behaviours only, leaving little material to engage with within the application that would provide forward-leaning motivation.

Fortune City is a personal finance application and game available on both the Android and iOS platforms. It consists of an RPG-style game in which the user's character builds a city filled with inhabitants. Daily visits to the application and expense entries contribute to rewards within the game. The application allows users to track past spending and it employs an envelope-style budgeting system. For the purposes of this paper, Fortune City will be considered a hybrid application, since it provides financial data to the user on only their past behaviours, making it passive and leaving little material with which to motivate future behavioural change. However, the application does include several active features which motivate the user to engage with the game portion of the application and thus may motivate users to return to the game. The personal finance portions and game portions of the application intersect superficially.

YNAB is a personal finance budgeting application. It is available on both the Android and iOS platforms. Every Dollar is another personal finance budgeting application. It is available only on the iOS platform. Both allow users to track their spending from the past and employ an envelope-style budgeting system as well. However, they are considered to be active for the purposes of this paper, because they both encourage users to allocate their earnings into the future. This allows users to actively engage with the application. Of note is that these two applications were the only two of the 51 applications reviewed that contained this feature.

6.1.3 Gamified Features

The following is a review of the gamified elements which have been implemented into the selected personal finance applications.

Money Manager's only gamified feature lies within budgets. In the application, each budget includes a reverse progress bar, which is drawn down as you spend money. Through the inclusion of this feature, Money Manager employs Chou's core drives of scarcity and avoidance as users try to avoid running out of budget before month's end. Within Cugelman's framework, the reverse progress bar motivates users to avoid punishment, the suggestion being that that mechanism will reinforce more positive spending behaviours. Though the extent to which users intuit or internalize this implied advice can't be determined.

Mint's gamified features lie exclusively in its data visualizations which provide post-spending feedback. These visualizations include a net worth graph, which allow users to feel a sense of accomplishment and ownership as they accumulate wealth or pay down debts. Users may experience scarcity and avoidance as they view Mint's reverse progress bars attached to each budget category. Mint's cash flow line graph may help users feel empowered as they receive live feedback on their performance. The above are the core drives Mint addresses from Chou's framework. Under Cugelman's architecture, Mint provides reinforcement in the form of the reverse-progress bar's indicator of remaining budget. The cash flow line graph provides live feedback.

Fortune City is the only application being reviewed that presents overtly as a game. Within Chou's framework, Fortune City creates a sense of meaning for users by creating the narrative that users are challenged to create an economically prosperous city inside the game. The game includes characters that each have their own statistics and story. Users develop a

sense of accomplishment through building their in-game city and receiving points and badges for their work. Every time users enter their real-life expenses, they are provided with positive feedback in the form of virtual diamonds which can be exchanged for perks inside the game. This fosters a sense of empowerment in the user. The game allows users to build their own city and there are a range of citizens and badges users can collect during play, which builds a sense of ownership. The game includes a number of social features like the ability to share accomplishments and photos with friends, and a leaderboard that can be viewed by connected players. Finally, the game includes three unpredictable elements: new visitors to the city, new public construction which takes place while the user does not have the game open, and a news feature which delivers fun facts about game characters. Under Cugelman's framework, Fortune City provides goal setting in the form of the story which challenges users to build a city and create economic prosperity. Including levels and challenging users to find new citizens to give them jobs helps users see their capacity to overcome challenges. The size of the user's city, their level and number of badges allow them to measure and compare their progress against their own and that of others. The social features allow for social connectivity and the characters and story make the experience both fun and whimsical. The real-life features of the game provide feedback through a highlight of frequent purchases, reverse-progress bars attached to budget categories, spending trend graphs, and breakdowns of spending on wants versus needs versus self-investments.

Every Dollar has some features which actively engage users within the application. First, the application prompts users to immediately budget every dollar they earn, when they earn it. There is a prompt under the "Planned" tab that indicates how much of a user's income is yet to

be budgeted. Once all of the income has been allocated, users receive a green check mark and the words “It’s an EveryDollar budget”. This feature, along with a line graphs which indicates Income vs. Spent provide users with a sense of accomplishment. During the onboarding process, the application leads users through a simple budget set up, at the end of which they are provided with a functioning budget. This help users feel a sense of empowerment as they figure out how to do a budget for perhaps the first time. While EveryDollar does not contain any social features itself, it does have a surrounding community. There is an active Reddit group named after EveryDollar’s creator Dave Ramsey in which EveryDollar is discussed, and Dave Ramsey publishes a call-in style podcast in which he shares listener stories and questions. Many of those callers are users of the EveryDollar product. That concludes the summary of gamified features from Chou’s framework. Under Cugelman’s architecture, EveryDollar promotes goal setting in the form of an emergency fund. Users are encouraged to build 3-6 months of savings that they may use as an emergency fund. Users overcome the challenge of onboarding and developing their first budget. The application provides continuous feedback in the form of a “Remaining” tab, which tells users how much remains in each budget category and how much remains for the month in total. The application also provides spending insights. EveryDollar provides some social connectivity in the form of the Dave Ramsey Reddit group and call-in podcast.

YNAB (short for “You Need A Budget”) is among the most gamified of the applications reviewed. Like EveryDollar, it challenges users to budget every dollar they earn, when they earn it. It goes further however, and challenges users to allocate enough money ahead of them to cover their next 30 days of expenses. Once users have accomplished that, they are challenged to continuously increase a metric called “Age of Money”, which is a loose representation of how

long the user's money has been in their account. Accomplishing these things provides users with a feeling of empowerment and prompts them to avoid losing ground. A running net worth tally creates a sense of ownership. Goal setting functions which include progress bars, a running net worth tally and red, yellow and green indicators to show when a budget has been satisfied all provide a sense of accomplishment to the user. YNAB's Age of Money indicator is an unpredictable element since it does not move in a clear and transparent way, but is an overall indicator of success. Users experience scarcity in the form of the income they have yet to earn. YNAB is at its most fun and active only on days when the user receives pay. It is at this time that users can satisfy budget categories that require filling. Success is indicated by the category number turning from orange to green. Finally, the last element of gamification under Chou's framework is YNAB's facilitation of social interaction. While the application itself does not include social features, YNAB has an active community on Reddit of over 120 000 users. User fandom extends to merchandise which includes apparel and home goods. Using Cugelman's architecture, YNAB facilitates goal setting through the desire it creates to have "all green" budgeting categories. Those are categories which have been adequately filled. Users are able to set up budget category goals and are challenged to set aside at least one month of expenses, then increase their age of money to over 30 days. Users build their capacity to overcome challenges as they go through the onboarding process, which is formidable in this application. They must also overcome the challenge of finding, then setting aside money to cover 30 days of expenses. YNAB provides feedback in the form of the green, orange and red indicators on budget categories that show whether the category has been filled, partially filled or overdrawn. It also

provides feedback through its net worth and age of money indicators. Last, YNAB indirectly provides social connectivity through its active and engaged following on Reddit.

The following tables show the incidence of each of the two expert's gamification characteristics in the five applications reviewed:

Table 1 - Analysis of applications through the lens of Yu-Kai Chou's Octalysis framework.

Core Drives	Number of Applications In Which It Appeared
Accomplishment	4
Empowerment	4
Social Influence	3
Avoidance	3
Unpredictability	2
Scarcity	2
Ownership	2
Meaning	1

Table 2 - Analysis of applications using Brian Cugelman's Behaviour Change Strategies.

Behaviour Change Strategies	Number of Applications In Which It Appeared
Goal Setting	3
Capacity To Overcome Challenges	3
Providing Feedback On Performance	3
Social Connectivity	2
Compare Progress	1
Reinforcement	1
Fun and Playfulness	1

These tables show that the core drives of accomplishment and empowerment have been well incorporated into modern mobile budgeting applications, as have the strategies of goal

setting, capacity to overcome challenges and providing feedback on performance. However, the core drives of meaning, ownership, scarcity and unpredictability have yet to be well incorporated. Likewise, the strategies of comparing progress, reinforcement and fun/playfulness have not been well incorporated.

6.1.4 Analysis

The above is a representation of the personal finance apps those interested in a more financially literate Canada have to work with. Three of the applications reviewed focused exclusively on providing feedback on past performance. This can leave consumers directionless as they try to dissect what went wrong and spend time feeling guilty about past financial failures. It also distracts consumers from the formidable task of future planning and provides them with no direction on exactly how to go about that, instead leaving it to the consumer to intuit what the most productive path might be. Mint, Fortune City and Money Manger have users plan only so far as they would need to to avoid overspending the current month's budget. All three failed to account for future planning in their designs.

While the features of Fortune City may pique a gamification enthusiast's interest within this group of contenders, it fell short in the area of fostering financial well being. Fortune City employed the "accounting of past performance" model of financial tracking and it failed to integrate most of its gamified features with the real-life financial ones. Public construction, the characters, story, level ups, social sharing, leaderboards and a host of other game elements had no link to the real financial performance of its users. Only real-life expense entry is linked to the application's gamified features in the form of diamonds (this game's digital currency), buildings

and badges, all of which are awarded upon the entry of an expense by the user. All other game mechanics within the game operate independently of the real-world financial behaviours of the user and so, apart from perhaps motivating the user to use the game, have little to no bearing on their level of financial wellness. Fortune City also offers the option to buy digital goods in the game using fiat currency. Things such as diamonds, themes and builders (the characters that build the user's city) are available for purchase. The game also has the option to watch video advertising in exchange for more diamonds. Both mechanics run counter to sound financial and time management principles. Fortune City can be credited with building the habit of expense entry in its users and it provides beautiful data visualizations which could aid users in synthesizing the feedback on past performance the application provides.

YNAB and EveryDollar have the ability to give users some choice in what they will do with their money in the future, giving them real personal agency to make change. EveryDollar and YNAB both encourage users to allocate all of their money to pre-determined budget categories immediately upon receipt. This provides users with currency "within the game" with which to make meaningful change in their lives. Both applications also include an element of education. EveryDollar does so through its figurehead Dave Ramsay and his numerous free and paid online resources. YNAB does so through a number of free and paid online resources purpose-built for the application's users. Both encourage users to pursue the basic financial literacy tenet of building an emergency fund, and provide users with a metric against which they may measure their progress. Both have managed to amass an outside following in which users of the applications congregate and engage online, independent from organizational efforts put forth by these businesses.

In addition to the aforementioned features, YNAB includes additional features that retain and engage the user. One of the most engaging features is the colour coding of budget categories. In the application, categories that have been overspent turn red, indicating an immediate need to resolve a budget shortfall. Categories that have been partially filled in pursuit of a budget deadline turn orange. Categories that have been fully satisfied turn green. During the allocation of incoming funds, these indicators give the user clear direction as to where the funds should go based on their previously inputted financial priorities. They also give the act of receiving a paycheque into YNAB a “whack-a-mole” quality as the user eliminates underfunded categories by assigning money to them until they all turn green. Users on the Reddit group r/ynab have attested to this feature’s addictive quality. YNAB also facilitates staged financial goals. The first stage is to become fully onboarded, which involves the user inputting their entire financial life into the application. The second stage is to start allocating funds to budgets with the aim of getting ahead of the budgets’ actual due dates. The third stage is to break free from living paycheque-to-paycheque by having enough money set aside to cover 30 days of expenses. After that, YNAB challenges users to increase their Age of Money to 30 days and beyond.

6.2 Recommendations

The market review highlighted gaps in the selected digital applications, most notably, there is an opportunity for the development of a simpler and more intuitive onboarding process to help get users start effectively using the application. This would create a mobile budgeting

application and financial literacy tool that is more accessible to Canadians who might otherwise feel intimidated or overwhelmed during the process of entering their full financial picture into an app. There is also an opportunity to make financial knowledge more available to users by leading them along a formal financial literacy skills-building path inside the app. This could start the user along the path towards F.I.R.E. by teaching them the basic tenets of the movement, such as mastery of short term habits while simultaneously building and maintaining an awareness of long-term goals. There is an opportunity to make it clear to users what they should do next to progress financially. Finally, there is an opportunity to help users overcome their cognitive biases around money by using rewards and other motivational affordances to help users adopt and employ more productive financial habits and behaviours.

All three new features could employ gamification to increase user motivation and engagement. Within the realm of fintech & gamification, it's clear that meaning, fun and playfulness have yet to be significantly incorporated into a financial literacy-focused personal finance app in Canada. It's also clear that fostering a sense of ownership in the user is an opportunity to improve the selection of applications on offer. There is also room to use scarcity, unpredictability and reinforcement in the design of a new budgeting application. Finally, increasing the user's social connectivity, and giving them the ability to compare their progress to that of others or of themselves is an opportunity to improve on the selection of applications currently available.

This paper proposes building a mobile budgeting application prototype which includes the beneficial existing features from the five applications reviewed, new onboarding features designed to get users using the app, skills building features designed to help the user get on the

path towards F.I.R.E., and habit-forming features that compel users to use the app on an ongoing basis. The end goal is to have users engage, through the use of gamification, in more financial wellness-building behaviours in their everyday life.

6.3 Prototype Design

6.3.1 Introduction

The framework outlined in this paper's recommendations will be embodied as a mobile budgeting application.

The application's three key features will be:

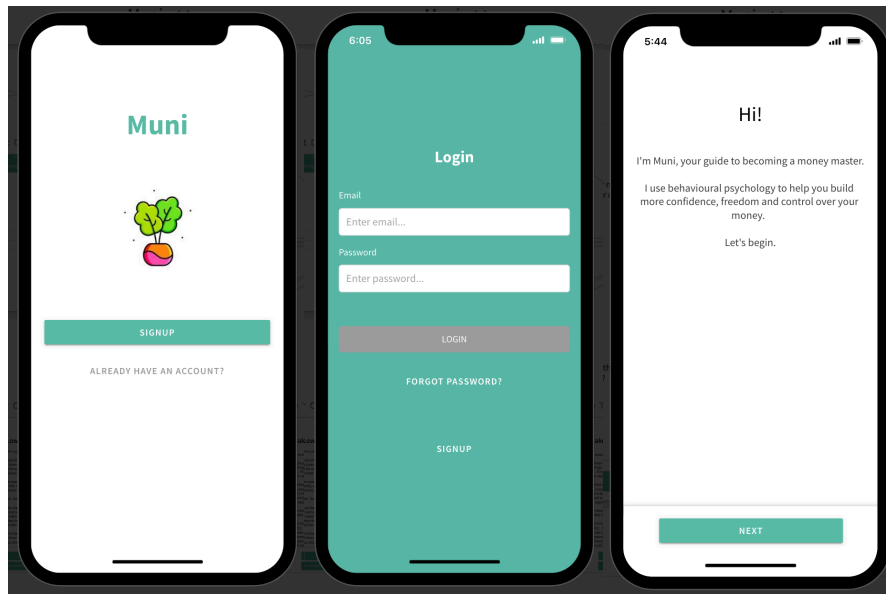
1. A simple, intuitive onboarding process.
2. The provision of a formal skills-building path towards greater financial literacy.
3. Rewards providing feedback to the user.

The inclusion of features will be supported by research. The gamification tenets employed within them are highlighted in bold and will be described in terms of Yu-Kai Chou's Octalysis framework. For a complete list of the app's features, see Appendix F.

6.3.2 Onboarding

The mobile app prototype begins with a robust onboarding deck designed to ease the user into the process of building healthy financial habits. The onboarding deck begins with the introduction of a human character named Muni.

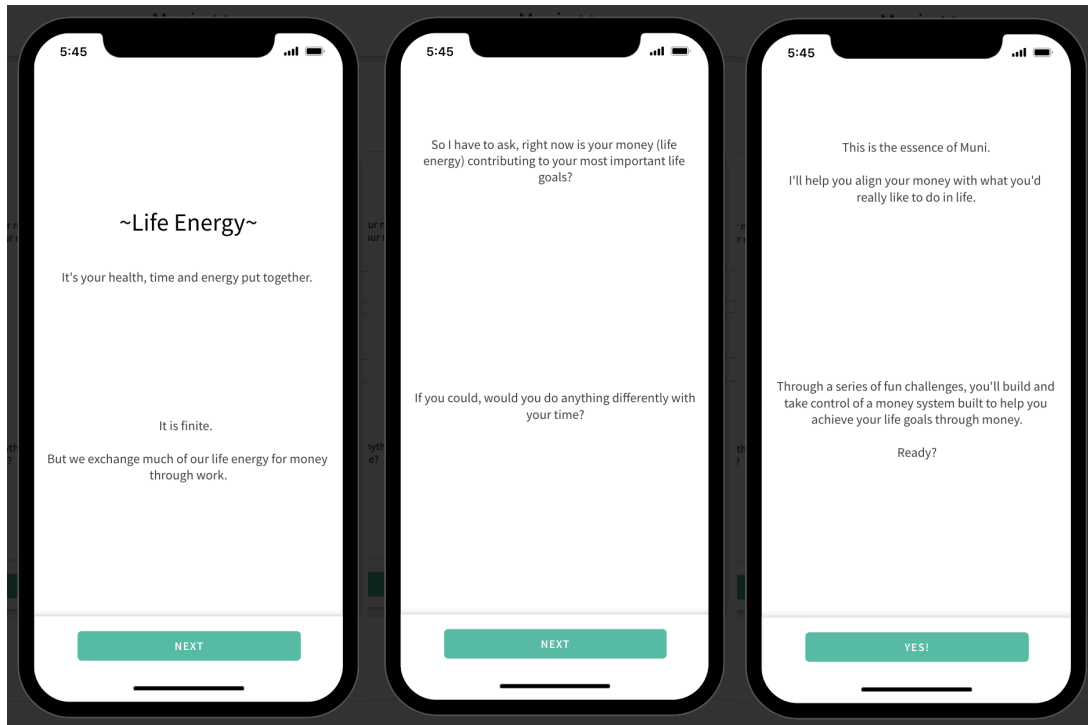
Figure 1 - Anthropomorphism of app in onboarding.



An **anthropomorphic character** was included because, according to Reeves & Nass (1996), social responses to certain types of computing systems are automatic and natural; human beings are hardwired to respond to cues in the environment, especially to things that seem alive in some way. A human voice delivered through digital media is seen by humans as a social actor. A social actor, according to Fogg (2002), can be persuasive by rewarding users with positive feedback, modelling a target behaviour and providing social support.

The onboarding deck then introduces the concept of “life energy” to the user. Life Energy is a concept borrowed from Vicki Robin’s book *Your Money Or Your Life* - the book that sparked the F.I.R.E. movement in the early 1990’s. The concept of Life Energy equates all money spent in the user’s life to pieces of their finite life span, or as Vicki puts it, “something you trade the hours of your life for (through work).” (Robin, 1992). The idea of Life Energy helps put how the user spends their money under a microscope.

Figure 2 - Introduction to life energy in onboarding.



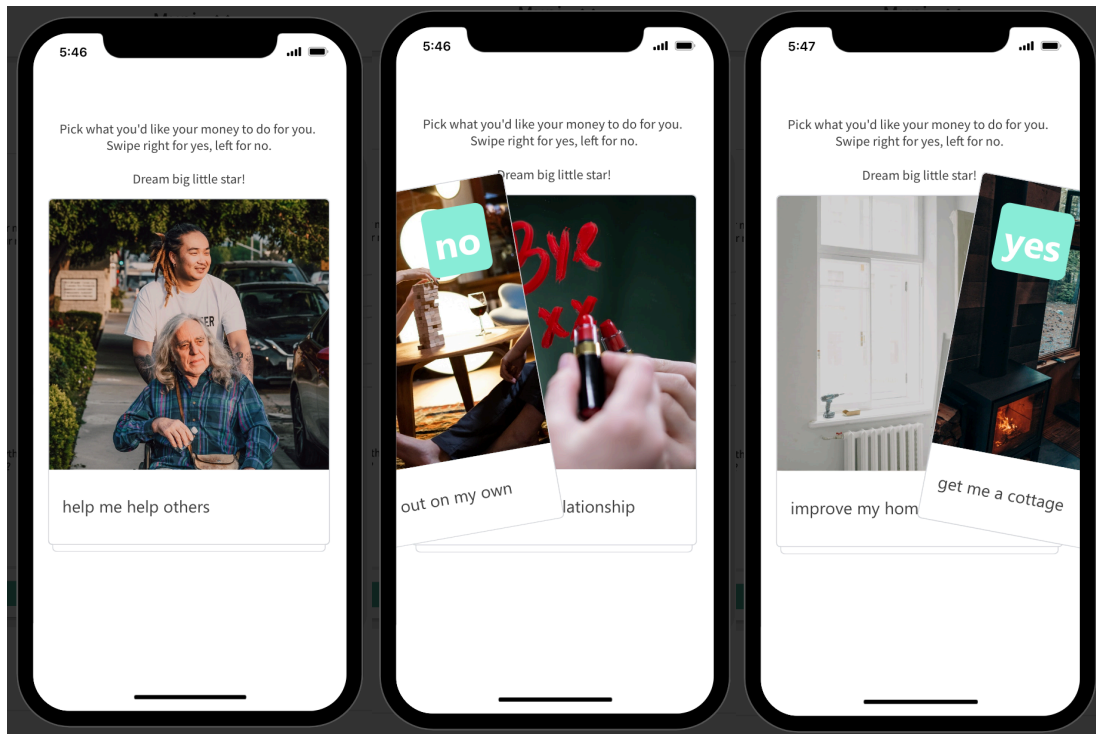
The onboarding deck then goes on to ask the user whether the life energy they spend is truly contributing to their most important life goals. This part of the onboarding deck introduces the game mechanic of Narrative, under the umbrella of Epic Meaning to the app which, according to Chou (2019) gives the user some context about why they should play the game. It adds a **deeper why** to the question of why the user should begin using the app. In a study done by a wearable technology company called Zamzee, the company gave children a fantasy quest in which users were provided with a narrative that had meaning, then asked them to run up and down real-life stairs a number of times. Zamzee showed that the kids who participated in the quest that contained a deeper meaning moved 59% more than the kids who were not provided with a quest filled with meaning. (Zamzee, 2012). This demonstrates that providing users with a

deeper why behind the ask that is being made of them, increases their motivation to do that work. The onboarding deck also employs the game mechanic FOMO Punch, under the umbrella of Loss & Avoidance to prompt the **fear of regret** in the user. The app does this by asking the user, “If you could, would you do anything differently with your time?”. This phrase was included to highlight that this app presents the possibility of time freedom for the user and frames it in such a way that the user would be walking away from that potential outcome if they stopped using the app, thus delivering the FOMO Punch described by Chou. Recent literature has demonstrated that fear arousal may motivate precautionary action (Ruiter et al. 2003; Rogers & Prentice-Dunn, 1997; Witte & Allen, 2000). This section of the onboarding was designed to motivate the user to use the application. It starts out with a simple philosophy (life energy) and follows quickly with questions designed to pique the user’s curiosity and deeper yearning for something different in life, that “something different” implicitly being a potential outcome of using the app.

Next, the app presents the user with a **card deck swiper** in which they can swipe right to say “yes” and swipe left to say “no” to various big-picture priorities. Studies have shown that making use of swipe gestures allows users to learn to use an app more quickly and easily. (Leitao, 2012; Loureiro & Rodrigues, 2011; Sears, Plaisant, & Shneiderman, 1990; Wigdor & Wixon, 2011). Users are asked to pick what they would like their money to do for them and they’re asked to dream big. This feature of the app employs the game mechanic of Evolved UI, under the umbrella of Scarcity & Impatience, which is a deliberate strategy used to make the onboarding of a game as **simple as possible**. Users who feel overwhelmed early in the use of an app, or feel that it will be too much work, tend to discontinue use of an app that presents said challenges. Evolved UI features are designed to make an application easy, non-threatening and

intuitive to use in the beginning. Added complexity is deliberately saved for later in the user flow.

Figure 3 - Card deck swiper used to define user's dreams.



Sony employed this when they launched a new onboarding process named Evolution UI. It was an intentionally hobbled Android launcher that forced the user to perform simpler tasks before the more complex features of the software were unlocked for users. (Cooper, 2014). Segway also employed this mechanic in their Ninebot personal transport products. New users of these products are asked to connect the scooter to an app on their phone. That application brings users through an onboarding deck called a New Rider Tutorial that discusses safety precautions one should take when using the device. Speed in the hardware is limited to 10km/h until the user has completed the New Rider Tutorial, thus preventing them from making high speed mistakes before becoming an educated user.

The card deck swiper employs the game mechanic of Dangling, under the umbrella of Scarcity & Impatience, by prompting the user to consider their most important life goals. By authentically and organically **dangling something of great importance** to the user during game flow, the application reframes the value of the game to be more valuable in the user's mind. The card deck swiper employed choice to the user's advantage, using the game mechanics of Choice Perception and Meaningful Choice, under the umbrella of Empowerment of Creativity & Feedback. Studies have shown that users like something more when they are provided with **options and choice**. In a study on the effects choice has on intrinsic motivation, results indicated that providing choice enhanced intrinsic motivation, effort, task performance, and perceived competence. (Patall, et al., 2018). When providing that choice, the app gives users choices that are meaningful and require careful consideration on the part of the user. It allows the user to create a playing experience for themselves within the game that is unique and customized, thus, according to Patall, increasing its perceived value in the user's mind.

Once the user has made a decision about all of the cards in the card deck swiper, they receive a pop-up message from the voice of Muni saying "Outstanding! If I've missed anything you can add it later.". This implements the game mechanic Mentorship, under the umbrella of Social Influence & Relatedness. Mentorship helps motivate users through the process of onboarding by **providing a supportive voice**. It also presents the game mechanic Beginner's Luck, under the umbrella of Epic Meaning, **making the user feel they have succeeded early in the game**, upon the completion of a relatively easy and pleasurable exercise.

Figure 4 - Anthropomorphism providing mentorship in onboarding.



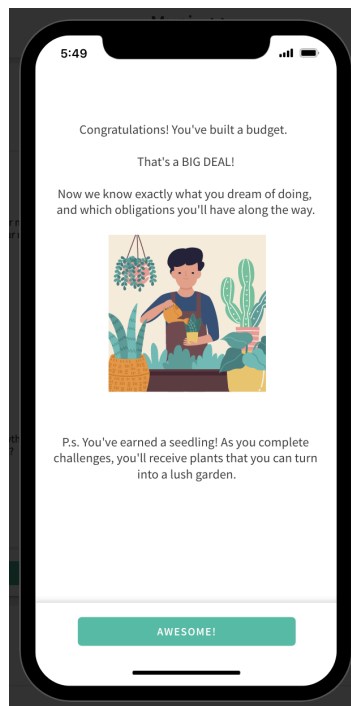
Next, users are provided with a card deck swiper that asks them to select the bills they have to pay in the next year from a set of pre-defined cards to further complete their list of budget categories. This simplifies what can otherwise be a stressful experience. Bill obligations are placed after dream items in the user flow so that the user is presented with a **positive daydreaming exercise**, before working on a more somber topic. Users are primed with the idea that using the app is about turning those daydreams into reality, before they have to grapple with thoughts of real bills, to prime them with positive emotions before the more neutral or negative exercise.

These two card deck swiping exercises also make use of the game mechanic The Alfred Effect, under the umbrella of Ownership & Possession. The Alfred Effect is when users feel that

a product is so **personalized to their own needs** that they cannot imagine using another service. (Chou, 2019). In a study done on the effect of personalization on a user's willingness to continue using a social networking site, findings showed that personalization influenced both satisfaction and perceived switching cost, which eventually lead to influencing continuance intention in the social networking site's use. (Park, 2014). The second card deck swiper also further employs Choice Perception in the onboarding process.

A this point in the game, the onboarding deck pauses to congratulate the user and to introduce another game mechanic. Muni explains that the user has been awarded a seedling that they can eventually turn into a beautiful garden within the game.

Figure 5 - Introduction of rewards in onboarding.

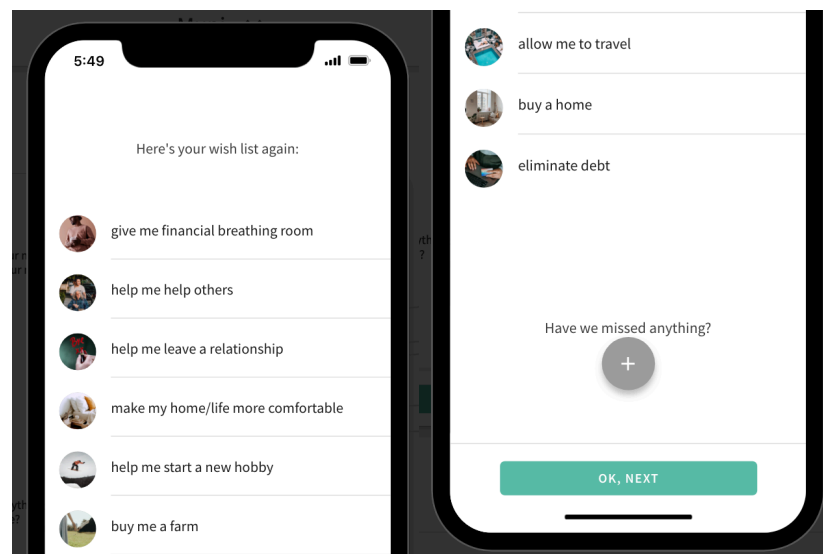


This provides an Achievement Symbol, under the umbrella of Development & Accomplishment, and introduces a Collection Set, under the umbrella of Ownership &

Possession. In this game, receiving rare and special plants that are awarded after the completion of hard work represent a **symbol of achievement** to the user and other players. The seedling awarded is part of a **collection of rare pixels**, in the form of plants, that the player can earn. Knowing that the seedling is one part of a total set motivates the user to “collect them all”. The inspiration for this feature came from the game, Pokemon Go.

The onboarding deck goes on to encourage the user to narrow their big goals down to their top three priorities so that they can focus on those and avoid feeling overwhelmed by the entire list they have chosen.

Figure 6 - Extreme customization feature in onboarding.



Users are then promoted to update their budget categories with customized names and the real monetary values it will take to satisfy them, further employing The Alfred Effect. The final steps encourages the user to set a date by which they would like to have accomplished the goal laid out in each budget category. The deck finishes with a Magnetic Cap, under the umbrella of Scarcity & Impatience. It limits the user’s engagement with the app to the first four skills

categories (of 17) that the app will cover. The app prompts the user to set the app aside for now and return for more easy challenges tomorrow. This mechanism contributes to the generation a flow state for the user. It **governs the user's interaction with the game** so that it's not too easy, but not too challenging. It also provides the user with the space required to process and integrate their experiences in the game. Researcher Csikszentmihalyi coined the term “flow” in his book *Flow: The Psychology of Optimal Experience*. In it he states, “The best moments usually occur when a person's body or mind is stretched to its limits in a voluntary effort to accomplish something difficult and worthwhile.” (Csikszentmihalyi, 1990).

6.3.3 Skills

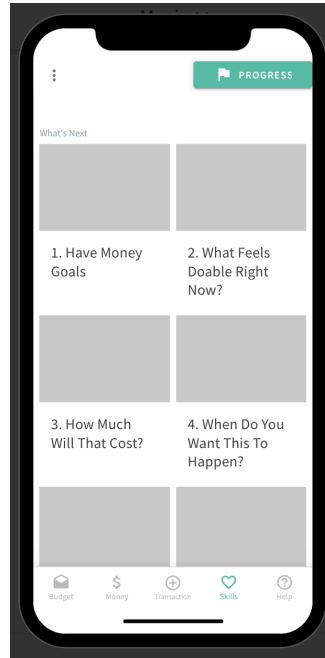
The skills section represents the second key feature of the app. It consists of a set of 17 financial literacy skills that will get the user onto the path towards F.I.R.E. These are the skills they will learn over the course of the game through the completion of **short challenges**. The skills encompass and expand on the financial literacy research and financial literacy education priorities identified by the Financial Consumer Agency of Canada. For a list and explanation of these priorities, see Appendix E. The 17 skills presented form a Collection Set, under the umbrella of Ownership & Possession that the user is challenged to complete as they move through the game. To examine the motivations behind the desire to collect, one can look at a 2012 study designed to reveal this propensity. In the study, when asked why they choose to collect, one respondent replied that they collected for “the challenge of the hunt; increasing and refining one's knowledge” and “having beautiful objects.” (Formanek, 2012). Collecting can motivate for these reasons.

The first five skills (of 17) in the app are visible the first time the user taps on the Skills tab within the app. By the time the user has entered the app interface proper, they have mastered the first four skills on their F.I.R.E. journey through the onboarding process. They are:

1. Have money goals
2. How much will that cost?
3. When you do want this to happen?
4. 4. What feels doable now?

These form a part of the user's Beginner's Luck, under the umbrella of Epic Meaning as they realize they have easily completed the first four challenges of the game. A 1975 study by Langer & Roth found that that early success, or beginner's luck, causes the user to make a skill attribution early in the game. After the attribution is made, outcomes inconsistent with it are not given much weight. An early, fairly consistent pattern of successes leads to skill attribution which in turn, lead subjects to expect future success. (Langer & Roth, 1975). This suggests that **allowing wins to feel easy**, early in the game will motivate users to continue with game play even after experiencing a setback or more difficult challenge. The fifth skill: 5. Become aware of what you earn and spend., is visible as the next step the user is meant to take on this journey.

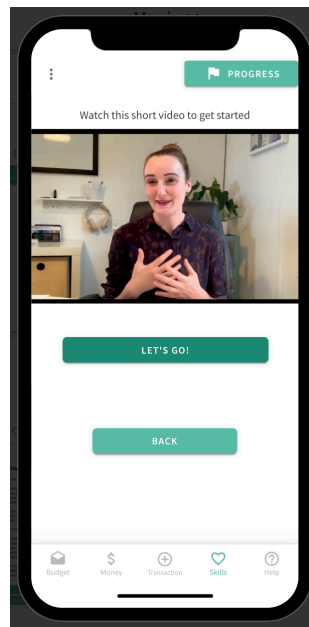
Figure 7 - Linear layout of skills development path.



The not yet fully revealed list of skills to be completed also employs the game mechanic **Glowing Choice**, under the umbrella of **Unpredictability & Curiosity**. It plays on user's curiosity, inviting them further into the game and **making it clear how they are meant to move forward** in the game. This satisfies what Jane McGonigal considers to be a requirement of a good game in her book *Reality Is Broken: Why Games Make Us Better and How They Can Change the World*. According to McGonigal, these games ensure players always know what to do next and provide an **abundance of meaningful, voluntary work** to the player. (McGonigal, 2011). The remaining 12 skills are hidden and **adaptively released** as Achievement Symbols under the umbrella of **Development & Accomplishment**. This mechanic makes having access to each new step an accomplishment for the user, since those new skills pages are not automatically nor immediately accessible. Access requires work (game play).

When a user clicks on a skill, they are taken to a page that gives them the option to watch an introduction video, jump into the challenge or leave. Here, users are given Choice Perception under the umbrella of Creativity & Feedback. Throughout their journey, the user remains in total control and is **given choices about how they would like to proceed**. Each skill begins with an introduction video in which a **real person contextualizes the challenge**, provides the user with three options for how to complete. The person in the introduction video encourages them to begin.

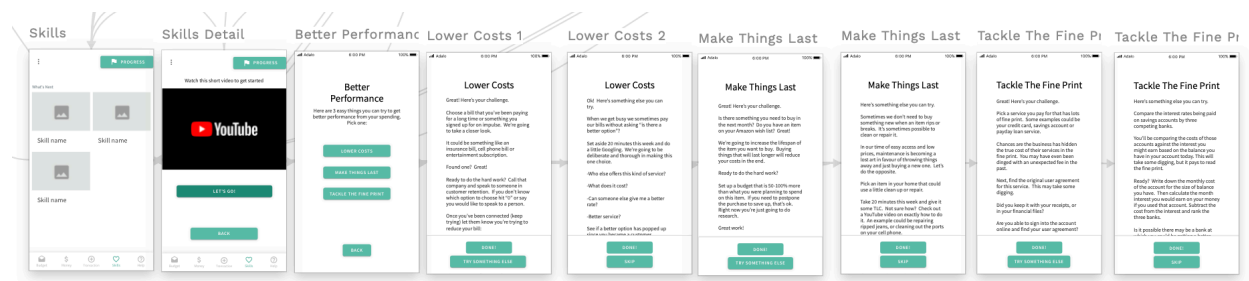
Figure 8 - Introduction page in skills area of the app.



This encouragement provides Mentorship under the umbrella of Social Influence & Relatedness. Providing numerous options for completing each challenge gives the user both Choice Perception and Meaningful choice under the umbrella of Creativity & Feedback. Users have a say in their personalized financial journey. They can choose which path feels best to them. This ensures every user will have a unique experience. Mentorship continues as a human

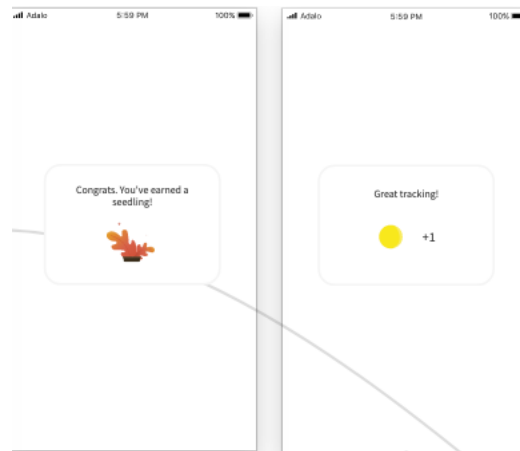
voice narrates the challenge in a written format. On each challenge page, the user has the option to quit the challenge or choose a new one, maintaining the perception of choice. The **narrated challenge** puts the user in the role of hero, as they fight against inequality through judicious use of their personal funds. In one narrative, *8. Achieve better performance*, the user works against businesses that take advantage of their disorganized finances to take back control of their own money. Both narratives make the user a Humanity Hero, under the umbrella of Epic Meaning. The game **brings a world mission to the user** through careful management of their own finances. As the user does so in this game, they contribute to something greater than themselves. The user is lead through a simple challenge related to the skill they have chosen. Once they have completed that challenge in real life, they confirm its completion within the game.

Figure 9 - User flow of a skills challenge which demonstrates user choice.



When the user has confirmed the completion of a real-life challenge, they are rewarded with a unique seedling for their garden. All 17 seedlings form a collection set that corresponds with the 17 skills users progress through in the game. The user earns sunlight and water to grow their seedling into a full-sized plant through daily earnings & expense entry.

Figure 10 - Seedling & sun reward modals within the app.



The game also releases a new skill the following day. The awarding of a seedling and new skills challenge uses the game mechanics of Achievements Symbols (under Development & Accomplishment) and Milestone Unlocks (under Creativity & Feedback) to motivate the user. Having the game delay the release of the next skill until the following day employs the game mechanic of Magnetic Caps, under the umbrella of Scarcity & Impatience, contributing to the generation of a flow state for the user. The game makes the action of completing a skill that is desired rewarding, but withholds commencement of the following skill by 12 hours. This also creates **artificial scarcity**. The scarcity of a desired game mechanic increases the user's motivation to return to game play.

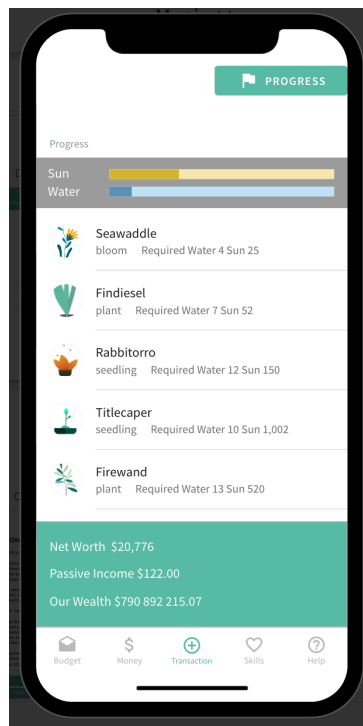
The skills area of the app introduces the first of three central engagement loops within the game. The rewards and positive encouragement the game provides motivates the user to want to progress and improve. They are prompted to take the action of completing a new skill, then upon completion, provided with immediate feedback in the form of a new seedling and potentially, better real-life financial performance as a result of acquiring that skill. The **positive feedback**

and rewards motivate the user to return to skills section of the app to participate in the engagement loop a second time.

6.3.4 Progress

The final key feature of the app is the Progress section. It takes the form of a single page that gives users a simple, easy to read overview of their real-life finances, as well as their progress in the game.

Figure 11 - Rewards page within the app.



It employs the game mechanic of Narrative (under Epic Meaning) through the continuation of the narrative introduced during onboarding, which is that the user is building a beautiful garden within the game. During this quest, they will be collecting a set of 17 seedlings

(Collection Sets, under Ownership & Possession). They are challenged to grow these seedlings into full-sized plants through the application of **sun and water points**.

Here, the first four seedlings the user has earned through their Beginner's Luck in onboarding (under Epic Meaning) are already shown. Each plant is shown in a list which in the game, represents an Achievement Symbol (under Development & Accomplishment) to the user. Their garden acts as a Trophy Shelf (under Social Influence & Relatedness) through which the user may display their achievements in an obvious way to others who are familiar with the game. Chou (2019). The garden draws the user into a Sunk Cost Prison, under the umbrella of Loss & Avoidance. A **sunk cost prison** compels the user to continue using a game, even if their motivation wanes, for fear of losing the time and effort they have put into building their trophies, which in this case, are the plants in their garden and their accurate **total net worth** number.

Above the list of seedling is a Progress Bar (under Development & Accomplishment). The **progress bars** shows the number of sun and water points the user has collected through their entries of income and expenses. Each time the user enters real-life income into the app, they are rewarded with water points. Each time the user enters an expense into the app, they are rewarded with sunlight points. These resources accrue on the user's Progress page. Sunlight and water represent Exchangeable Points (under Ownership & Possession) in the game. A study done by Ueyama et al. (2014) established that the inclusion of top level gamification mechanisms like points and badges in an activity increased the participation probability from 53% (without gamification) to 73% in participants. Users use sunlight and water points to feed their seedlings so that they may grow. Each plants starts as a seedling, progresses to a mid-sized plant, then a full-sized plant through the application of these exchangeable points. This

encourages users to use the points they have earned in a strategic and scarce manner, motivating them to stay engaged in the game. (Chou, 2019). The **final evolution of each plant is kept a secret**, playing on the user's curiosity. Each plant has a unique requirement for sunlight and water before it will grow. Some plants require more sunlight, and therefore income recordings from the user. Other plants require more water, or expense transaction recordings. Users are provided with Meaningful Choices (Creativity & Feedback) as they strategically decide how to allocate the sunlight and water points they have earned. Users are able to **choose which plants to grow and when**.

The excitement of this process is enhanced by the **randomizing of these rewards** in the game. This employs the game mechanic of Random Rewards, under the umbrella of Unpredictability & Curiosity. As users enter income and expense transactions into the app, they are awarded a randomized number of sunlight and water points. According to Chou (2019), this recreates the excitement that children have on Christmas Eve by not allowing the user to find out what the true nature of the award will be until a desired action has been performed. The game also provides Easter Eggs, under the umbrella of Unpredictability & Curiosity. Easter eggs are **sudden surprise awards** that are given out during the game without the prior knowledge of the user. This kind of unexpected positive event creates excitement in the user and prompts them to share that event with others. According to Chou (2019), it may even cause the user to speculate about which game actions caused the easter egg to appear.

Below the garden is the user's net worth, **year-to-date passive income** and the **collective net worth of all players in the game**. These figures provide **feedback to the user on their progress in real life** and develop a Monitor Attachment (under Ownership & Possession) in the

user. As the user sees their net worth and passive income increase (or decrease) they will feel compelled to return to the game to monitor those numbers. This may also apply to the achievements they have made in their garden. A study by Kirk & Swain (2018) supports this mechanic, providing evidence for the potential of psychological ownership emerging in relation to digital objects despite their intangible nature. (Koles & Nagy, 2021).

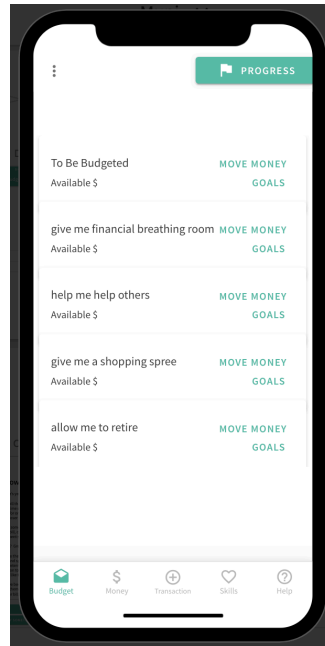
The garden mechanic encourages users in their daily habit of income and expense tracking, promoting a micro view of the user's finances. The net worth and passive income numbers encourage the user to take a macro view of their finances, striking a balance between short term and long term financial thinking. This addresses the cognitive bias of Discounting the Future (see Appendix 1). The **collective net worth** number represents the total net worth of all of the game's current users. It brings the individual user into a Group Quest (under Social Influence & Relatedness), in which they are working with a larger group of users to battle inequality. This feature allows users to feel they are making a greater difference and through the management of their personal finances, are contributing to a greater good in the world. In her book, *Reality Is Broken*, Jane McGonigal highlighted that World of Warcraft's focus on collective effort within the game is part of what makes the game so compelling to users. The game tabulates the total hours of play and displays it to users. As of 2020, World of Warcraft remains the world's most played video game, with 10.4 million years of collective play time. (Moore, 2020). This game mechanic also links users to an overarching game Narrative and makes the user a Humanity Hero as they work together with a larger group of users to solve a problem and make a difference. Users, without sharing their personal financial information, can feel they are working on a team towards a collective goal. This mechanic may also act as an

indirect way of facilitating real-life conversations about money between users. A study done on the efficacy of an e-learning game that used game mechanics to introduce users to a quest and encouraged them to participation in a group effort within the game suggested this as a possible outcome. In the study, students were challenged to improve their English speaking skills as they worked together with other players in the game. Students reported the game provided opportunities for them to make friends, express their opinions, and share their life interests with others. (Zheng et al., 2009).

6.3.5 Additional Sections

Budget categories that include a **goals features and making use of progress bars** are useful to the user because they make use of Progress Bars (under Development & Accomplishment) to increase user motivation. These features were present in several of the reviewed applications. Goals and progress bars also foster a Monitor Attachment (under Ownership & Possession) in the user as the user is motivated to return to the app to monitor their progress. **Highly customizable budget categories and goals** employ the Alfred Effect (of Ownership & Possession), making the app uniquely tailored to the user, such that they would find it difficult to move to another application.

Figure 12 - Customizable budget categories.



The customization factor within goals and budgets allows the user to gradually introduce **limitless complexity** to their budgeting app to more finely control their financial life. This employs the game mechanic of Evolved UI (of Scarcity & Impatience). Users are able to onboard simply and easily, then evolve the app to suit much more sophisticated financial circumstances as the user becomes proficient in the use of the app.

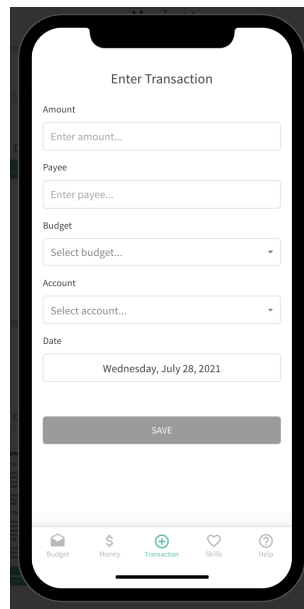
Having users put all income into a “To Be Allocated” category, then asking them to allocate all of those funds to budgeting categories the day it’s received converts the user’s real-life income into Exchangeable Points (under Ownership & Possession) in the game. Users can use their income to fill up goal and budgeting categories within the game, and as they do so, those categories turn from red (overdrawn) to yellow (underfunded) to green (fully funded). This colour coding of budget categories provides immediate positive feedback to the user. Having to

make choices about how any money received will be used in pursuit of the user's life goals, just as it's received, provides the user with Meaningful Choice (under Creativity & Feedback) and links that income, and choices regarding how to allocate that income, to the broader narrative within the game (more conscious use of life energy). It has the benefit of **speaking for all funds received immediately**, so that the user is never in a position of feeling that they have excess funds they can spend indiscriminately.

The "To Be Allocated" feature represents the second central engagement loop of the game. The act of earning income prompts users to enter their income into the app. When the user does this, they are provided with immediate positive feedback as they are able to fill incomplete budgeting categories that turn from yellow to green as they are filled. Users are awarded water points they can apply to growing their seedlings. The goal progress bars become increasingly complete as funds are allocated to the respective budget category. Finally, the net worth number increases over the short, and possibly long term. These four mechanisms for providing positive feedback make the act of entering income exciting, and may lead to the user feeling positive anticipation.

Additional features were designed for the budget and transaction entry areas of the app to increase user motivation and engagement. Under skill, 5. *Become aware of what you earn and spend*, users are encouraged to "**make an appointment with their money**" to build a habit of daily transaction entry.

Figure 13 - Transaction entry screen.



This feature implements an Appointment Dynamic, under the umbrella of Scarcity & Impatience. According to Chou (2019), this utilizes a formerly declared, or recurring schedule where users have to take desired actions to effectively reach a win state. In the case of this app, that is the desirable state the user has of having awareness and control over their money. The “daily appointment with their money” creates a prompt for action, which according to Fogg (2019), is one part of three in creating lasting behavioural change. Pleasurable engagement loops such as this one, in which the user feels motivation, acts, then receives immediate feedback can, over time, can become addictive to the user. The other two elements to creating this behavioural change are developing the right motivation (created during the app’s narrative and onboarding) and providing the user with the ability to the complete the action (created through the app’s easy to use UX/UI and its Skills feature). Should the user fail to enter any transactions on a give day, they are sent a **push reminder from the app to “check in with your money”**. This enacts the game mechanic Status Quo Sloth, under the umbrella of Loss & Avoidance. This creates and

external trigger that prompts the user to enter their transactions on a daily basis. (Chou, 2019).

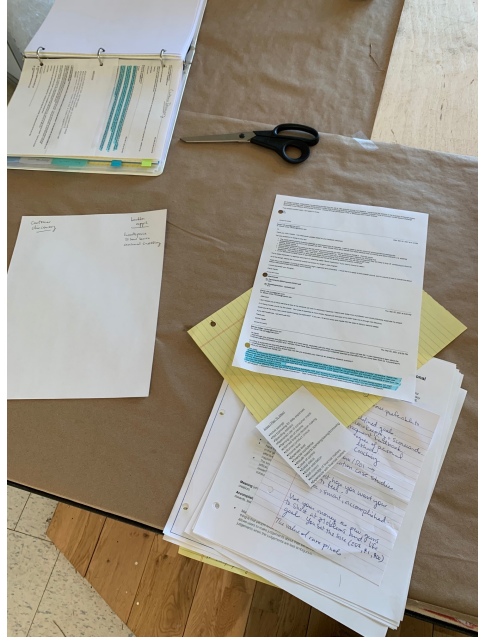
When the user does this, they are rewarded with sun and water points in the game and sometimes a randomized reward, which can be of relatively high value. As users enter their daily transactions, they are making time and emotional investments in the app, and building value for themselves. This encompasses the third and final engagement loop built into the app.

The final game mechanic employed within the app is Sunk Cost Prison (under Loss & Avoidance), in which users are motivated to continue to using the app to avoid losing perfect control of their money, since any lapse in transaction entry would result in an inaccurate picture of the user's finances. If more than 48 hours has passed since a transaction entry, the user is promoted with a **push notification saying “do you really want to lose perfect control of your money?”**. This is a strategy employed to help users overcome inertia on days during which they find it more difficult to track their finances.

6.4 Prototype Development

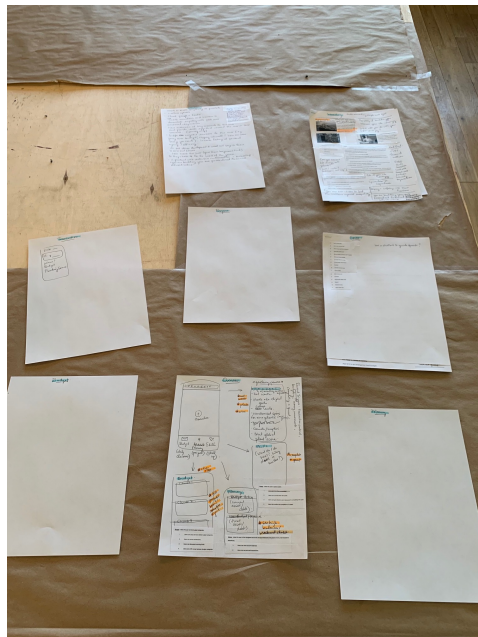
The development of the mobile budgeting application prototype began with a thorough review of research notes. Notes and inspirations were culled for content that was no longer relevant or useful to the prototype development process.

Figure 14 - Research notes.



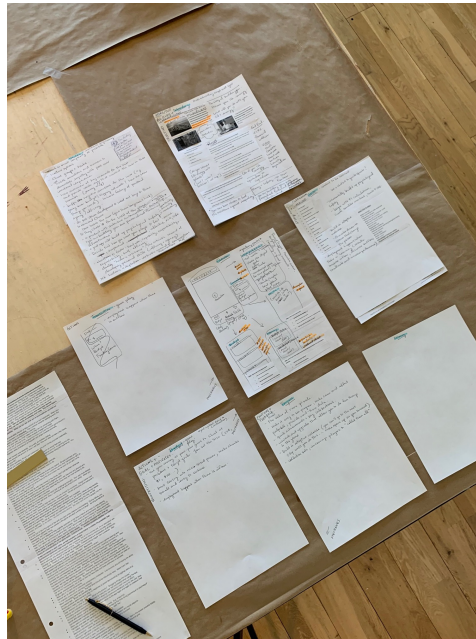
A rough paper prototype of the application was drafted to establish its basic layout. The budget, skills, progress, transactions and money pages are shown in the figure below.

Figure 15 - Paper Prototype of Mobile Budgeting Application.



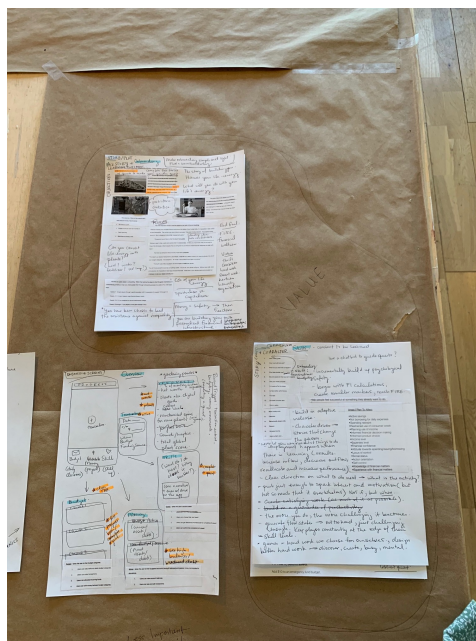
Research notes were grouped into themes aligned with the pages established in the app layout. Notes were reviewed to identify and eliminate duplication.

Figure 16 - Research notes grouped by theme.



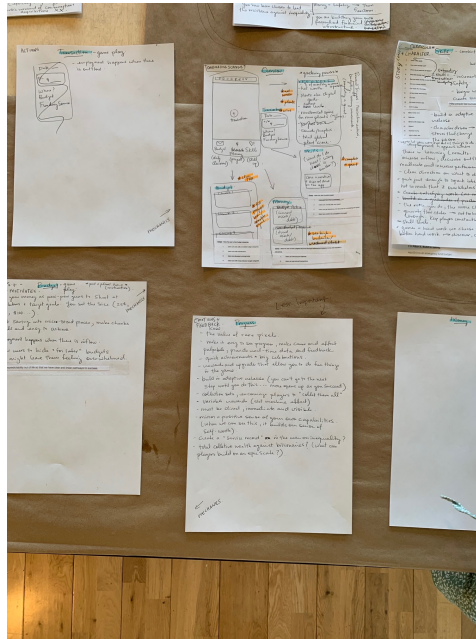
Areas within the app with the greatest motivational value to the user were then identified.

Figure 17 - Areas of greatest motivational value identified.



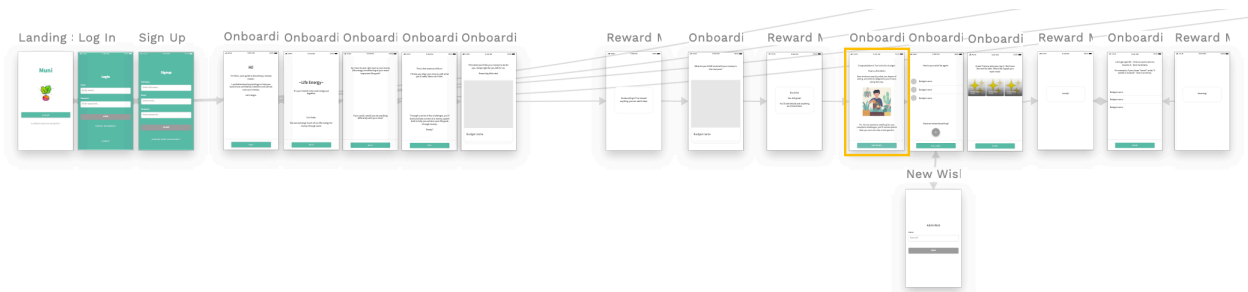
Likewise, areas within the app with the least motivational value to users were identified.

Figure 18 - Areas of least motivational value identified.



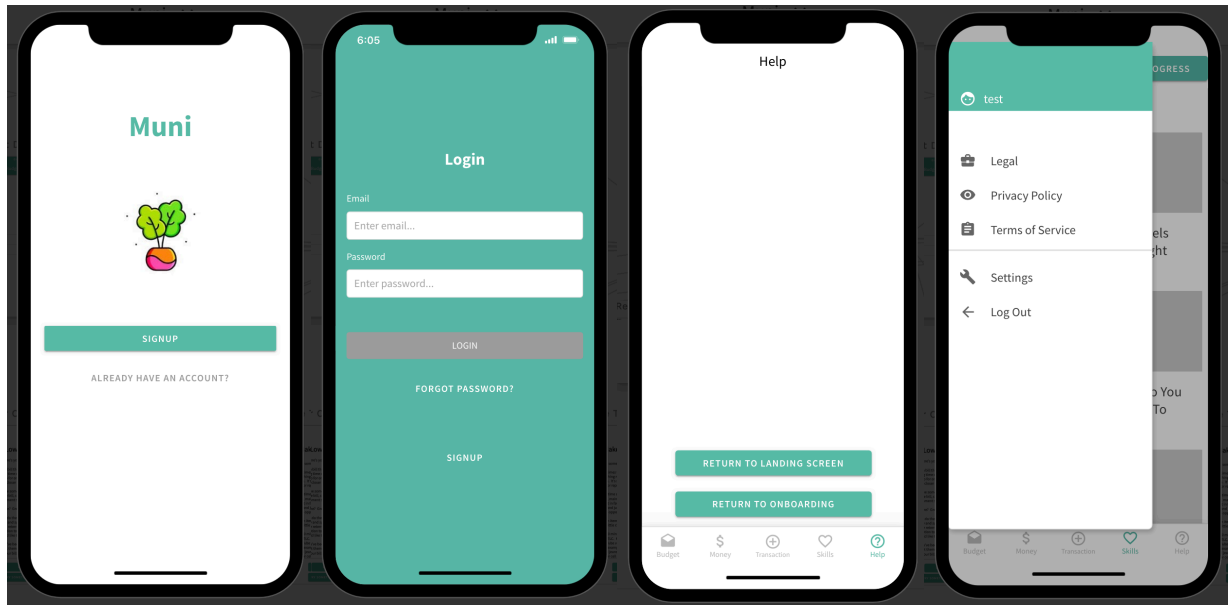
Following the thorough organization of research notes, the basic pages of the application were drafted out in the low-code mobile app development platform, Adalo.

Figure 19 - Low-code draft of application's basic pages.



Sign up, login and help pages were built, along with left and bottom navigation bars.

Figure 20 - Sign up, login, help pages & navigation.



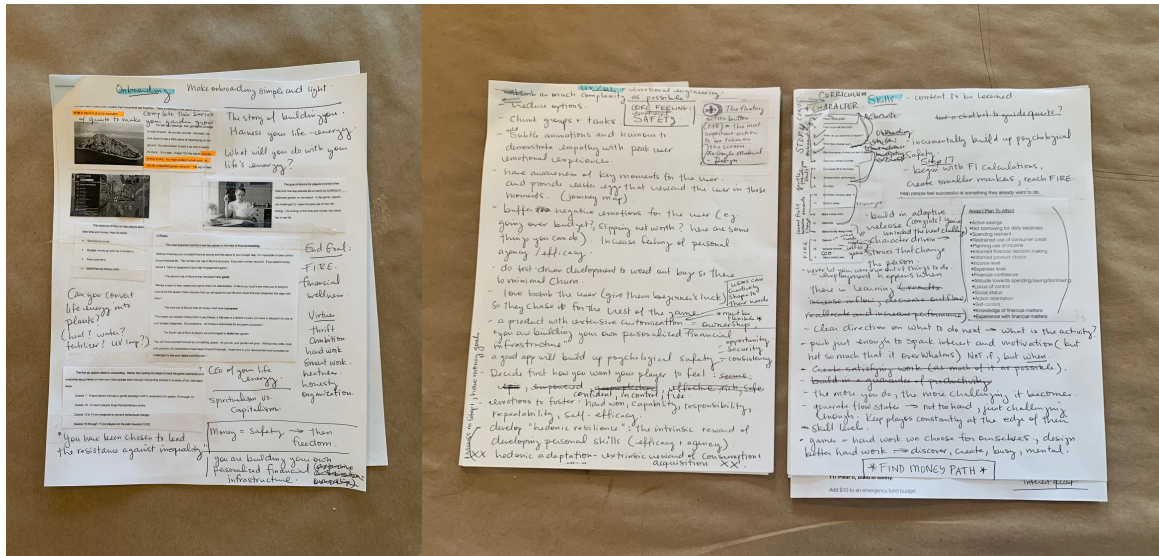
Next, the databases that would allow the app to function were designed and implemented.

Figure 21 - Application database design.

Database Collections		
Database Collections		
> Users	2 Records	
> Rewards	7 Records	
> Transactions	6 Records	
> Help Topics	0 Records	
> Payees	0 Records	
> Skills	17 Records	
> Budgets	50 Records	
> Accounts	3 Records	

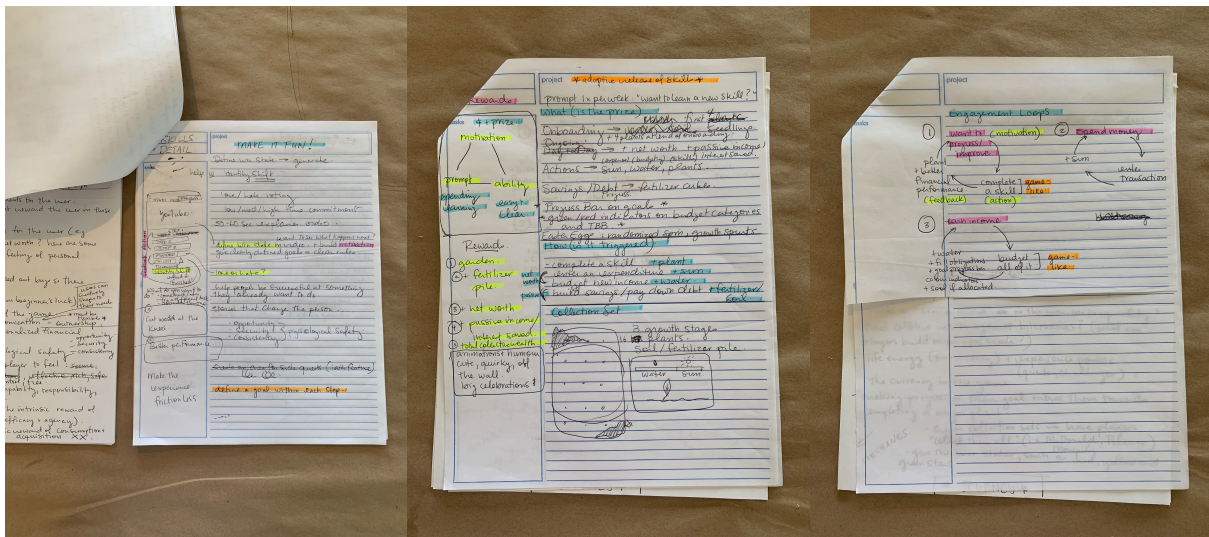
Research notes for each app page were reviewed and synthesized in greater detail to prepare for the creation of the app's final page designs.

Figure 22 - Review & synthesis of research notes for individual pages.



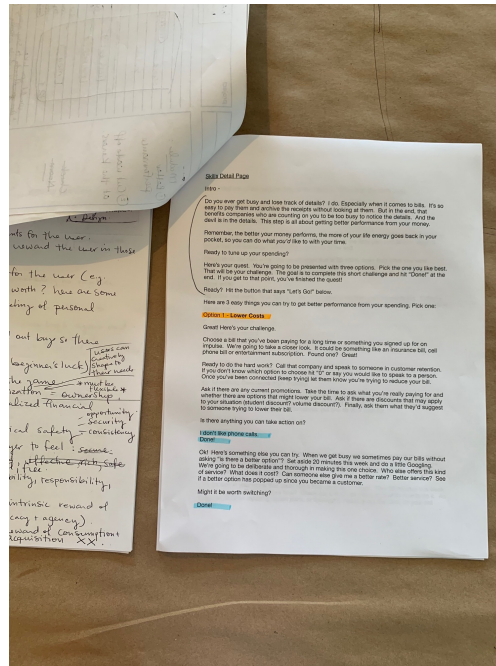
Research notes were synthesized into final drawings of the app's pages and gamified features.

Figure 23 - Final sketches of app pages & gamified features.



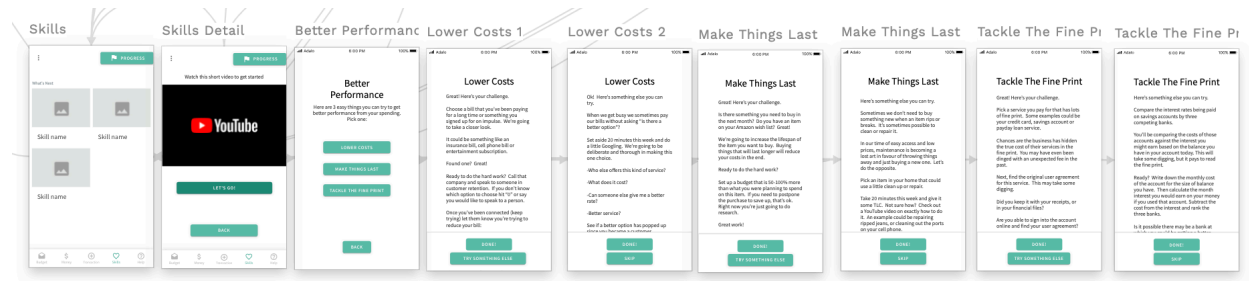
All narratives to be embedded in the app were typed in a word processing document, reviewed and edited before being included.

Figure 24 - Draft of game narratives.



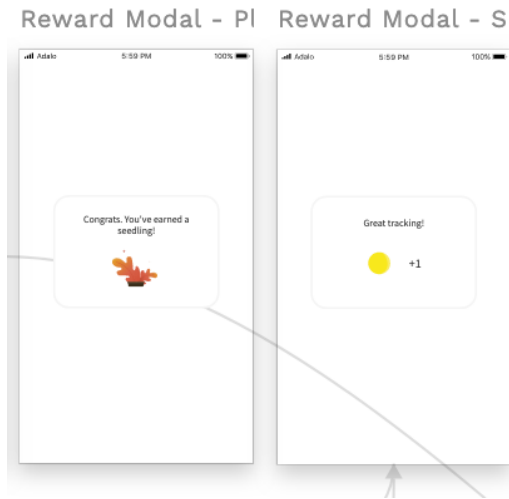
The game narrative was then fleshed out in Adalo through the drafting of appropriate pages and transitions for the written content.

Figure 25 - Game narrative draft on low-code platform.



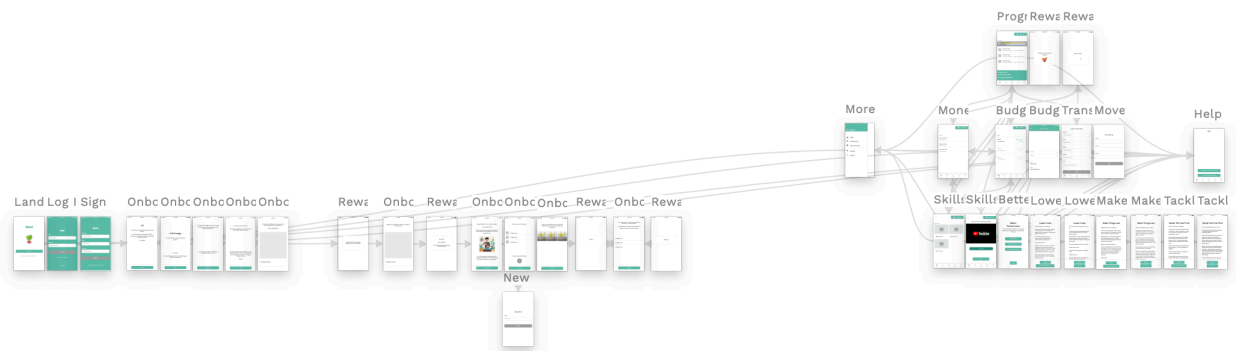
During this stage of the project, the app's databases were tested and reviewed to ensure the smooth functioning of the app. Individual rewards modals were constructed for inclusion and integration.

Figure 26 - Rewards modals.



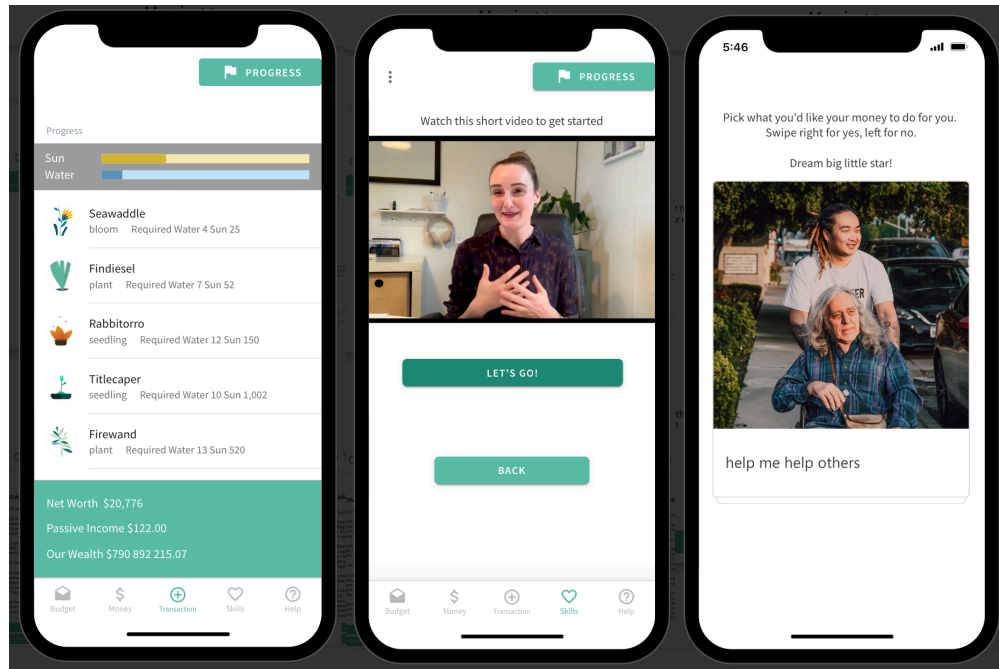
Interactivity between the app's pages was introduced and a pleasant and familiar user flow was developed.

Figure 27 - Application user flow.



The app's multimedia elements - including images, videos and animations were add to the app.

Figure 28 - Application multimedia elements.



Finally, the user flow and app functioning was tested to complete development of the prototype.

6.5 Takeaways

This application presents opportunities for users to simply and easily onboard into a financial literacy-focused budgeting app that increases in complexity with use. It makes financial knowledge and skills more accessible and available to users. Finally, it presents an opportunity to help users overcome cognitive biases to implement healthier financial habits and behaviours.

7.0 Conclusions

7.1 Conclusions

Prior to this project, the fintech industry had built the underpinnings of an incredible tool for financial literacy education that, this paper posited, it had yet to full utilize. For Canadians eager to offset the effects of rising inequality, it is hoped that this project has created a tool that will allow Canadians to feel a sense of personal agency, and increase their level of financial wellness.

This budgeting app prototype fills a gap in the Canadian mobile personal finance app market by employing the well-studied, but as of yet to be implemented game mechanics of meaning, fun, playfulness, ownership, scarcity, unpredictability, social connectivity, reinforcement and ability to compare progress. Through these game mechanics, and the implementation of behavioural change strategies and addiction-forming engagement loops, this project has created a mechanism by which users might make lasting and meaningful change to their level of financial well being.

7.2 Future Work

Upon conclusion of this study, there is room to do research on the relative efficacy of specific gamification techniques as defined by Yu-Kai Chou, Brian Cugelman and other gamification experts. Though it falls outside of the scope of this project, research of this nature may help establish which gamification techniques are most, and least effective when applied to improving the financial outcomes of users. To constrain the scope of this project, each

gamification technique was treated as having equal value, however further study might illuminate ways in which gamification might be optimally employed.

There is also room to do research on the real-life financial outcomes of users as a result of using the app prototype created in this study. This research would contribute to establishing the efficacy of gamification techniques when used in combination, and as applied to the financial wellness of users. This has the potential to act as a case study for the creation of more effective financial literacy education programs.

Appendices

Appendix A - Cognitive Biases

- Discounting The Future - the tendency we all have to give more importance to needs and wants in the present over those in the distant future.
- Overconfidence - being excessively optimistic and downplaying the possibility of negative events that could happen.
- Anchoring - our tendency to use the first number or other piece of information we encounter as our baseline for comparison.
- Confirmation Bias - the tendency to give more weight to new information or an opinion that confirms what we already believe to be true, and to ignore contradictory evidence.
- Loss Aversion - our propensity to feel regret over a financial loss more acutely than pleasure at a gain of the same amount (Menard, 2017).^[P]_{SEP}]

Appendix B - Excluded Mobile Budgeting Applications

Budgeting applications from the Google Play store that were reviewed, but not selected for discussion:

- Monthly Budget Planner
- 1Money
- My Budget Book
- AndroMoney
- Expense Manager Pro
- PocketGuard
- Wagestream
- Piggy Goals
- Daily Pocket
- Money Lover

Budgeting applications from the Apple App store that were reviewed, but not selected for discussion:

- Famzoo
- Wallet App
- Balance Viewer
- Monefy
- Spendee
- PocketSmith

- Mvelopes
- Pennies
- Our Expenses
- GoodBudget
- Fudget
- Buddy
- Daily Budget Original
- Emma
- Home Budget^[P]_{SEP}

Appendix C - Yu-Kai Chou's Octalysis Framework

In his book Actionable Gamification, Yu-Kai Chou categorizes gamified features into eight core drives:

1. Meaning - doing something greater than ourselves (narrative, elitism, hero, revealed heart, destiny, creation)
2. Accomplishment - making progress, developing skills, overcoming challenges (points, badges, rewards, leaderboard, progress bar, high five, crowning, boss fights)
3. Empowerment - a creative process of figuring things out, seeing results and receiving feedback (milestone unlocks, real-time control, instant feedback, boosters)
4. Ownership - we want to make what we own better, we want to accumulate more (exchangeable points, virtual goods, collection sets, build from scratch, avatar, protection)
5. Social Influence - mentorship, acceptance, social responses, companionship, competition and envy, seeing others do something that drives us, wanting to be closer to others drives us, we respond to nostalgia (friending, gifting, group quests, group flags, brag button, water cooler, conformity anchors, mentorship, social proof)
6. Scarcity - wanting something because you can't have it, it motivates us to think about it all day, make it exclusive and people will want to get in (appointment dynamics, magnetic caps, dangling, prize pacing, options pacing, countdown timer, torture breaks, moats)
7. Unpredictability - wanting to find out what will happen next, it is the primary factor behind gambling addictions, it plays on our curiosity (glowing choice, mini quests, visual storytelling, easter eggs, random rewards, obvious wonder, rolling rewards, sudden rewards)

8. Avoidance - trying to prevent something negative from happening, it's why we don't want to lose all of our work by quitting, or why we jump at time limited opportunities (sunk cost prison, progress loss, rightful heritage, status quo sloth, scarlet letter, visual grave, FOMO punch)

(Chou, 2019).<sup>[P]
[SEP]</sup>

Appendix D - Brian Cugelman's Persuasive Architecture of Gamification

Brian Cugelman categorizes gamified features into seven behaviour change strategies:

1. Goal Setting - committing to achieve a goal (providing clear goals)
2. Capacity To Overcome Challenges - growth, learning and development (offering a challenge, using levels)
3. Providing Feedback On Performance - receiving constant feedback through the experience (allocating points, showing progress, providing feedback)
4. Reinforcement - gaining rewards, avoiding punishment (giving rewards)
5. Compare Progress - monitoring progress with self and others (providing badges for achievements)
6. Social Connectivity - interacting with other people (showing the game leaders)
7. Fun and Playfulness - playing out an alternative reality (giving a story or theme)

(Cugelman, 2013).^[15]

Appendix E - Financial Consumer Agency of Canada's Research Priorities

In its 2020 report on the current state of financial literacy research in Canada, the Financial Consumer Agency of Canada highlighted the following research areas as priorities for the development of financial literacy curriculum:

- 1) Managing debt
- 2) Navigating the financial marketplace
- 3) Building savings
- 4) Budgeting

Managing debt involves the borrowing and repayment of money. Many consumers are unaware of the long-term costs of mortgages, credit cards, and other loans, and too few have a plan in place for paying them back.

Navigating the financial marketplace speaks to the growing complexity of the financial marketplace and to the consumer's ability to find the most suitable financial products and services.

Building savings involves putting money aside for the future, with some of the most common uses being to pay for emergency expenses, education, and retirement.

Budgeting is a fundamental component of financial planning and involves tracking income vs. expenses. As Canadians incur debt, the financial marketplace becomes more complex and harder to navigate, and with people saving less money overall, budgeting can help consumers meet their personal needs today and into the future.

(Canada Financial Consumer Agency of Canada, 2020).

Appendix F - Complete List of Prototype Application's Features

The application's features are documented include the following list. Features marked with an asterisk are concepts that have been borrowed from existing personal finance budgeting applications.

Key Feature 1: A simple, intuitive onboarding process.

Onboarding Section Game Mechanics:

1. Anthropomorphic character used in onboarding.
2. Provide a deeper why, through concept of "life energy".
3. Create a fear of regret by illustrating what the user will lose by quitting.
4. Use of card deck swiper.
5. Onboarding is made as simple and easy as possible.
6. Dangle something of great importance by having users think of their most important goals.
7. Provide options and choice in onboarding.
8. Providing a supportive voice and encouragement.
9. Makes the user feel they have succeeded early in the game.
10. Challenges happen only after a positive daydreaming exercise.
11. Have users personalize app to their own needs.
12. Provide symbols of achievement.
13. Introduce a collection or rare pixels to the user.
14. Govern or limit the user's interaction with the game.

Key Feature 2: The provision of a skills-building path towards greater financial literacy.

Skills Section Game Mechanics:

15. Provide short challenges.
16. Allow wins in the game to feel easy.
17. Make it clear to users how they are supposed to move forward.
18. Provide an abundance of meaningful, voluntary work.
19. Adaptively release skills challenges.
20. Give users choice about how they would like to proceed.
21. Use video of a real person to contextualize challenges.
22. Each challenge is narrated in a human voice.
23. Bring a world mission, as well as a personal mission to each challenge.
24. Create artificial scarcity through the withholding new challenges for 12 hours.
25. Provide immediate positive feedback through the award of a seedling.

Key Feature 3: Rewards providing feedback to the user.

Progress Section Game Mechanics:

26. Sun and water points awarded upon transaction entry.
27. Sunk cost prison in the form of a growing garden.
28. Total level of net worth.
29. Progress bars showing sun & water points.
30. Ability to allocate sun & water points to seedlings to choose which ones grow and when.
31. Final evolved form of each plant is kept a secret.

- 32. Randomization of levels of sun & water points awarded.
- 33. Sudden, surprise awards.
- 34. Year-to-date passive income.
- 35. Collective net worth of all players in the game.
- 36. Real-life numbers provide feedback to users on their progress in real life.

Additional Features

Budget, Money & Transactions Game Mechanics:

- 37. Goals feature that uses progress bars.*
- 38. Highly customizable budget categories & goals.*
- 39. Limitless complexity.*
- 40. Speaking for all funds received immediately.*
- 41. Ask users to make an appointment with their money.
- 42. Push reminder from app to “check in with your money”.
- 43. Push notification saying “do you really want to lose 100% control of your money?”

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